

VeriTrust

End-to-End Detection Flows and Hugging Face Model Map

Frozen architecture implementation reference

AUTHORITATIVE BOUNDARY: VeriTrust performs orchestration, validation, evidence, calibration and deterministic policy. All learned production inference runs on qualified Hugging Face-managed infrastructure. Third-party Inference Providers are **DISABLED_BY_DEFAULT**.

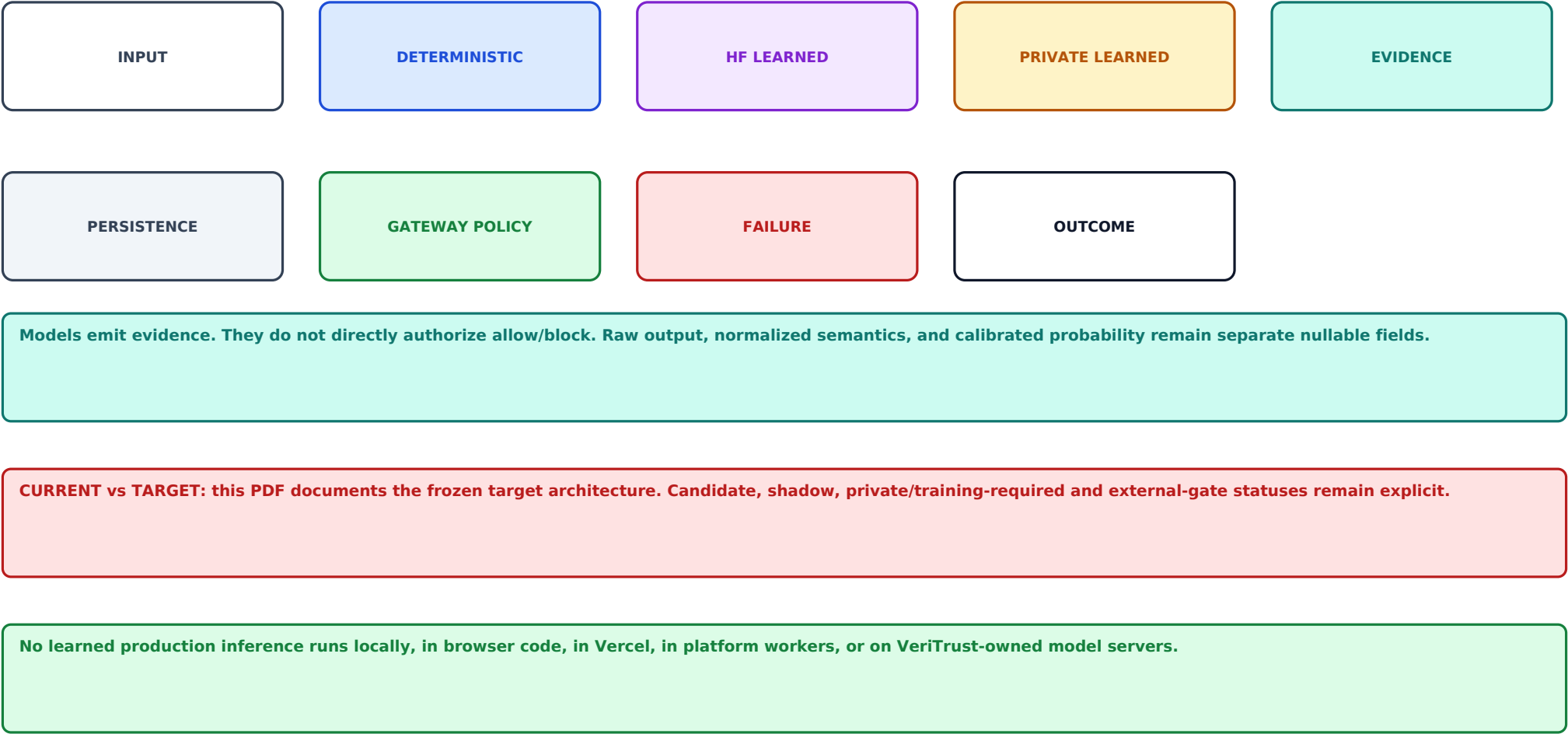
Canonical roles: 18 | Lifecycle states: 16 | Receipt: inference_receipt.v2

Generated only from the frozen Research package; no operational qualification result is implied.

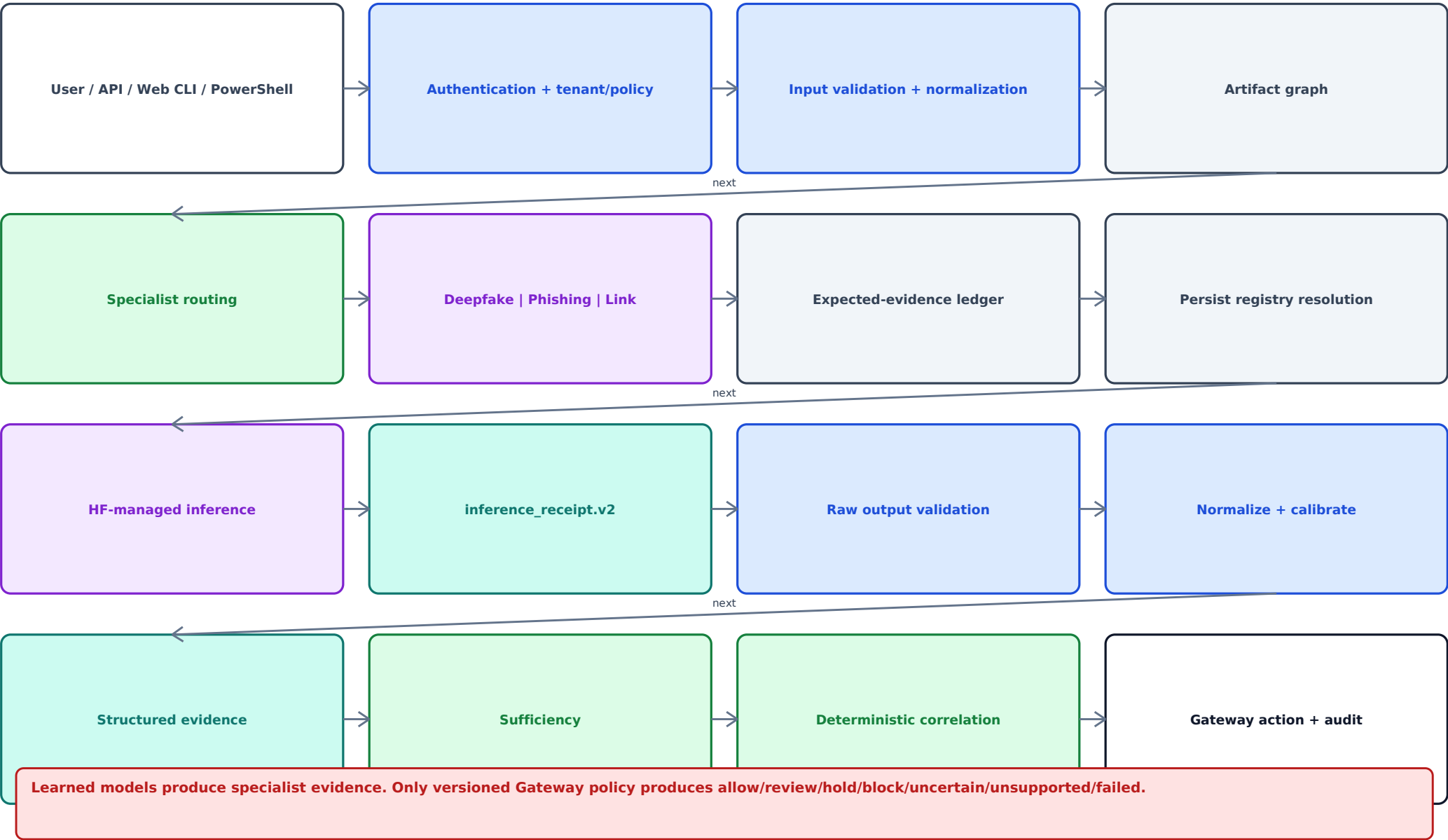
Executive visual summary



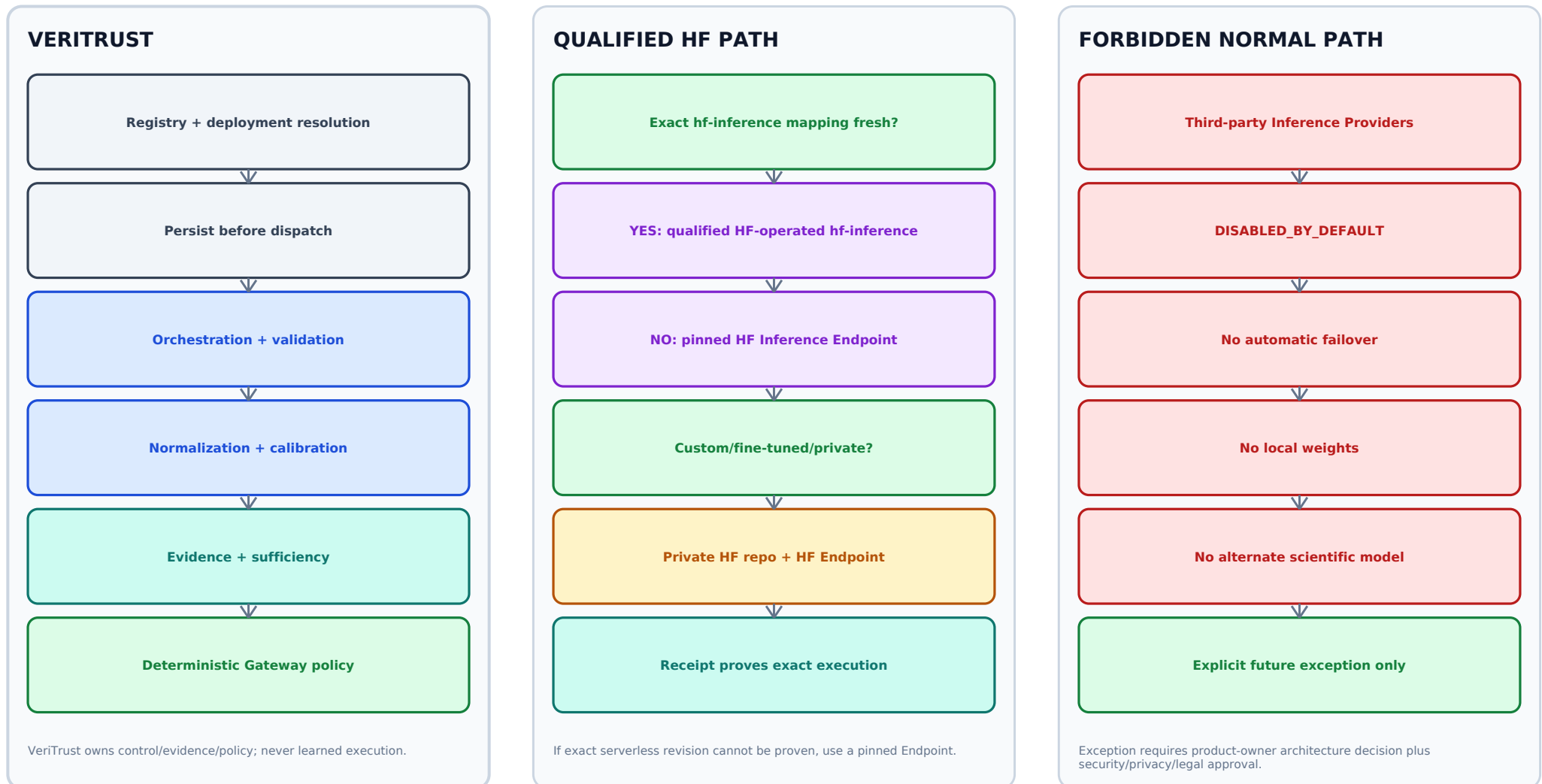
Diagram legend and non-negotiable boundaries



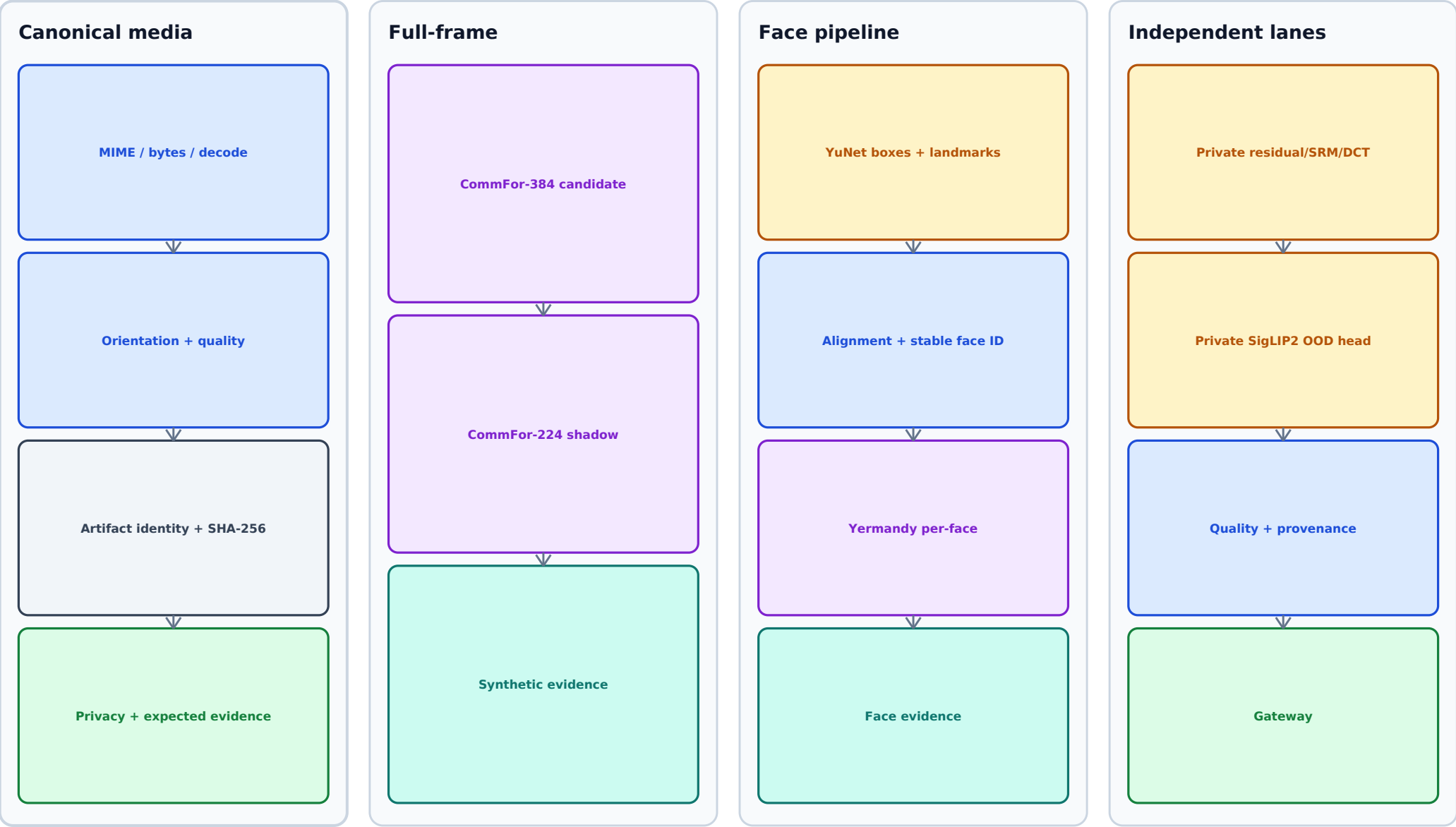
Global VeriTrust detection flow



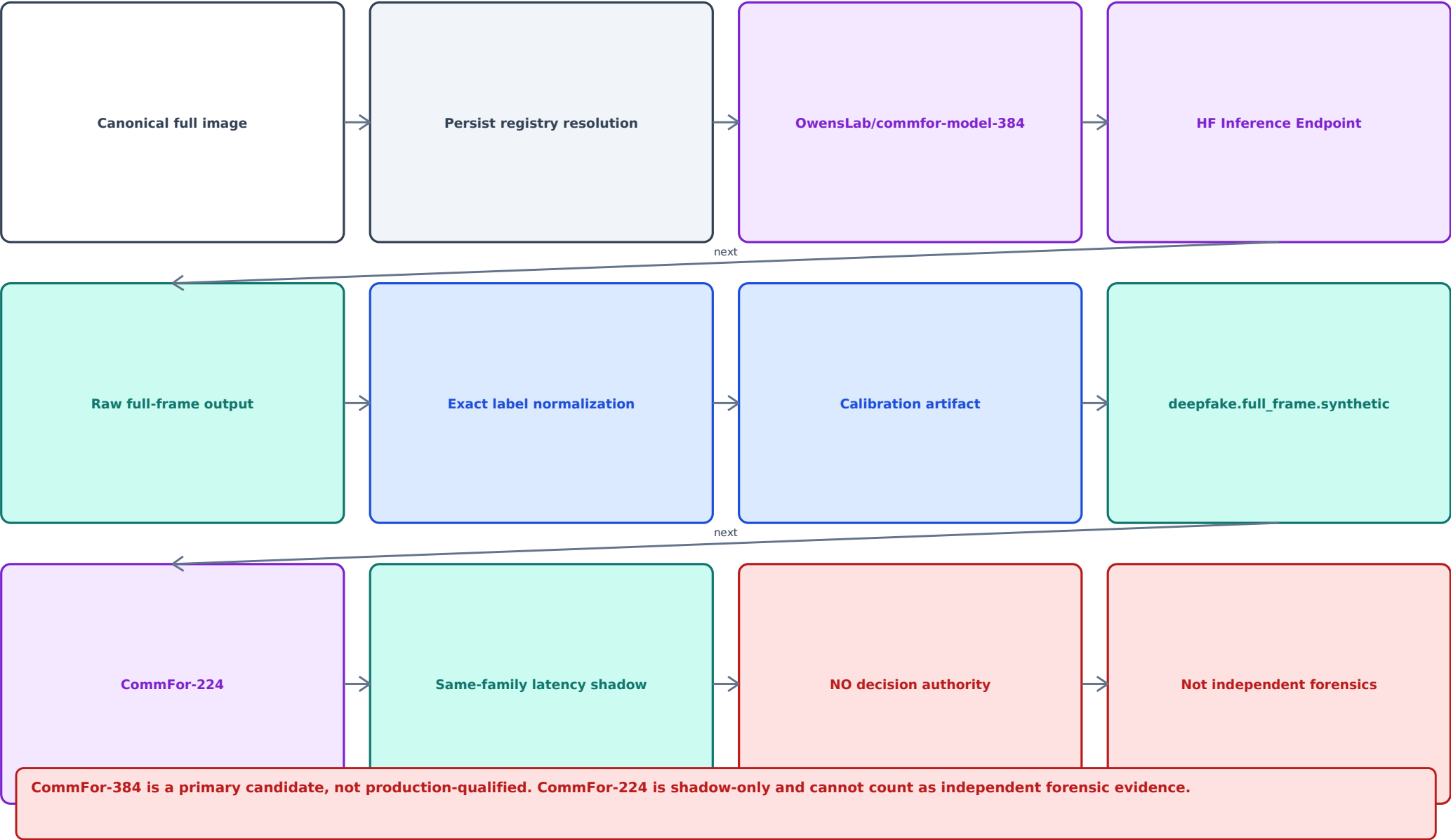
Hugging Face-managed execution boundary



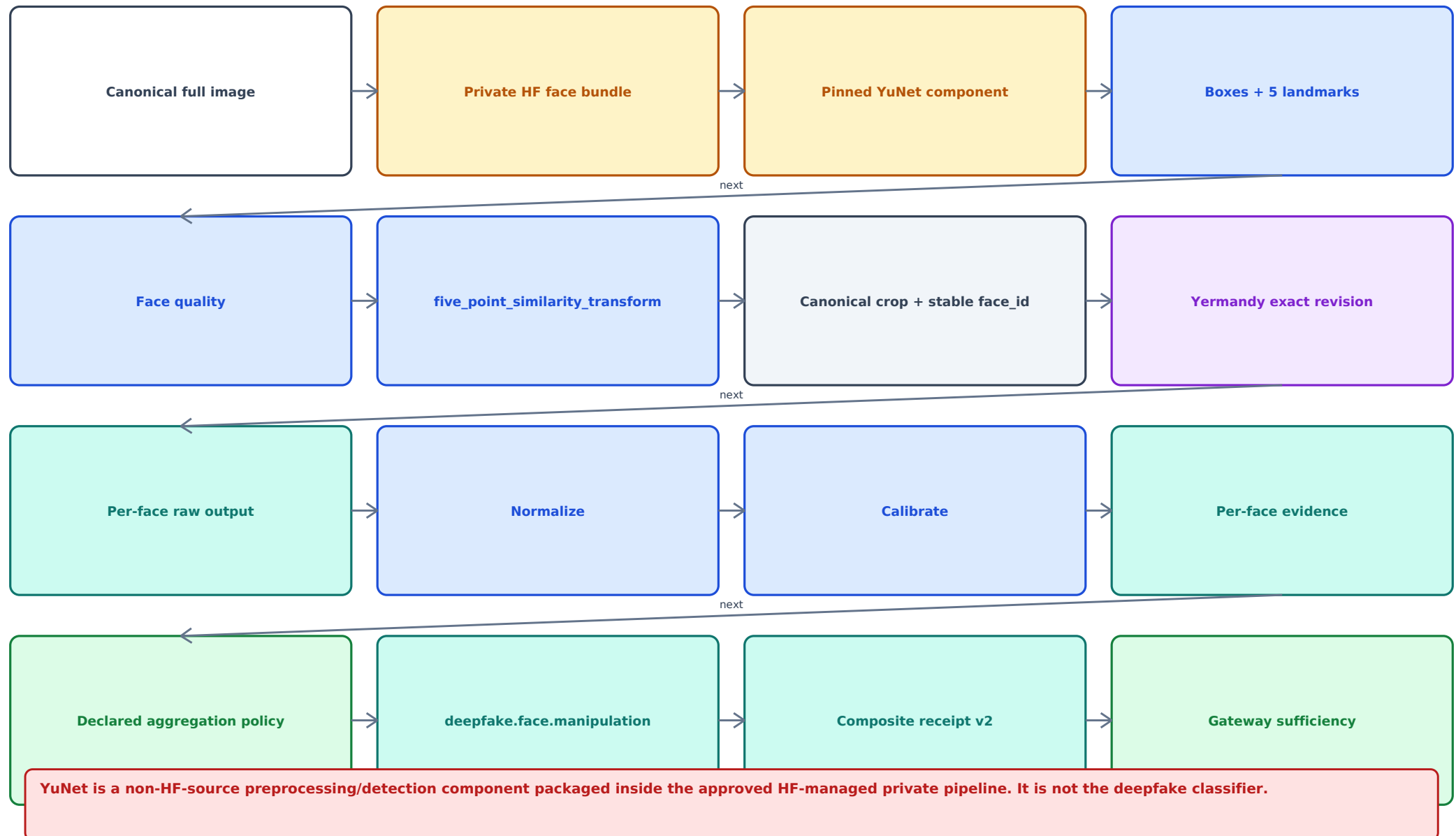
Deepfake: complete overview



Deepfake lane A: full-frame synthetic evidence

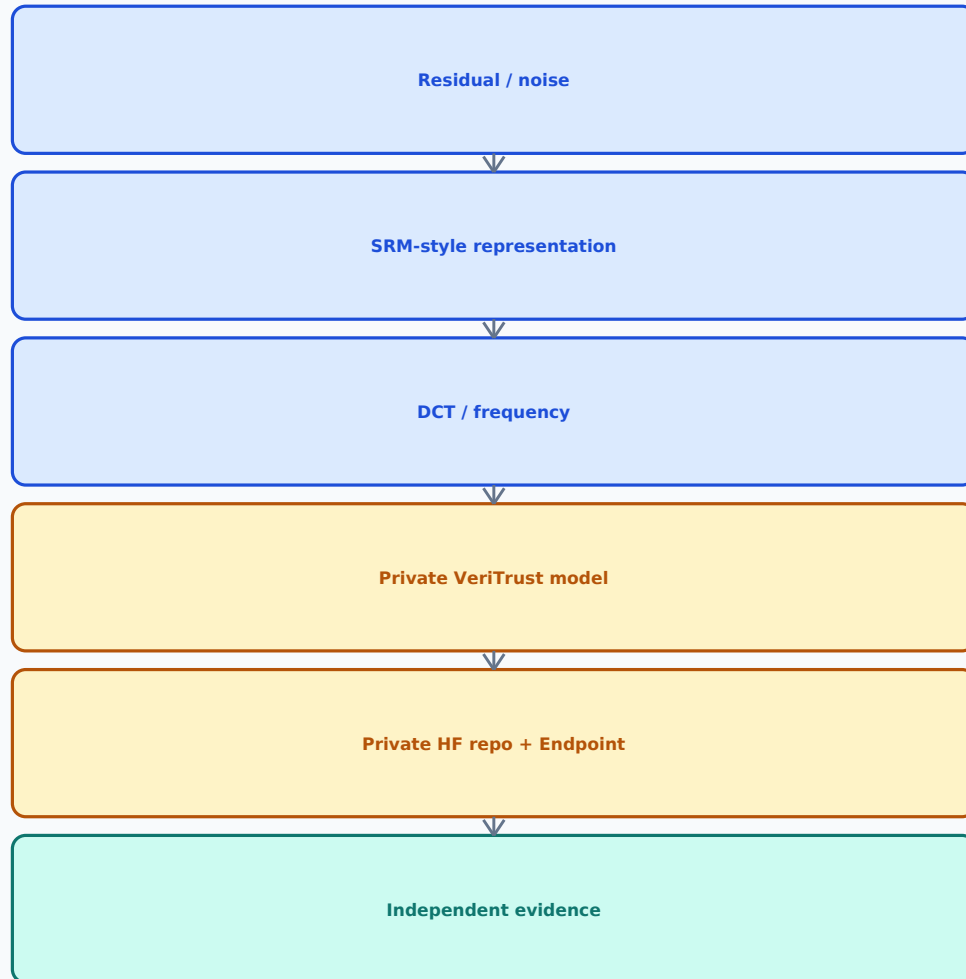


Deepfake lane B: face manipulation pipeline



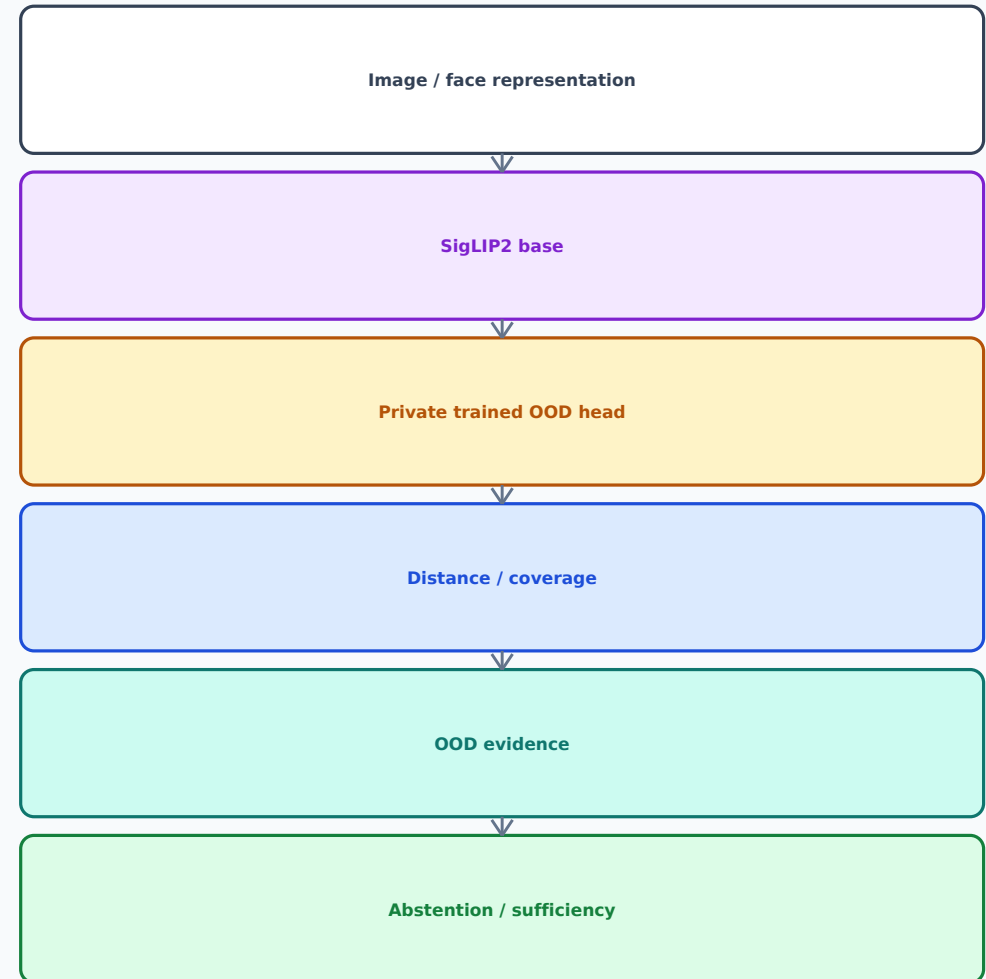
Deepfake lanes C and D: forensic independence + OOD

Independent forensic model



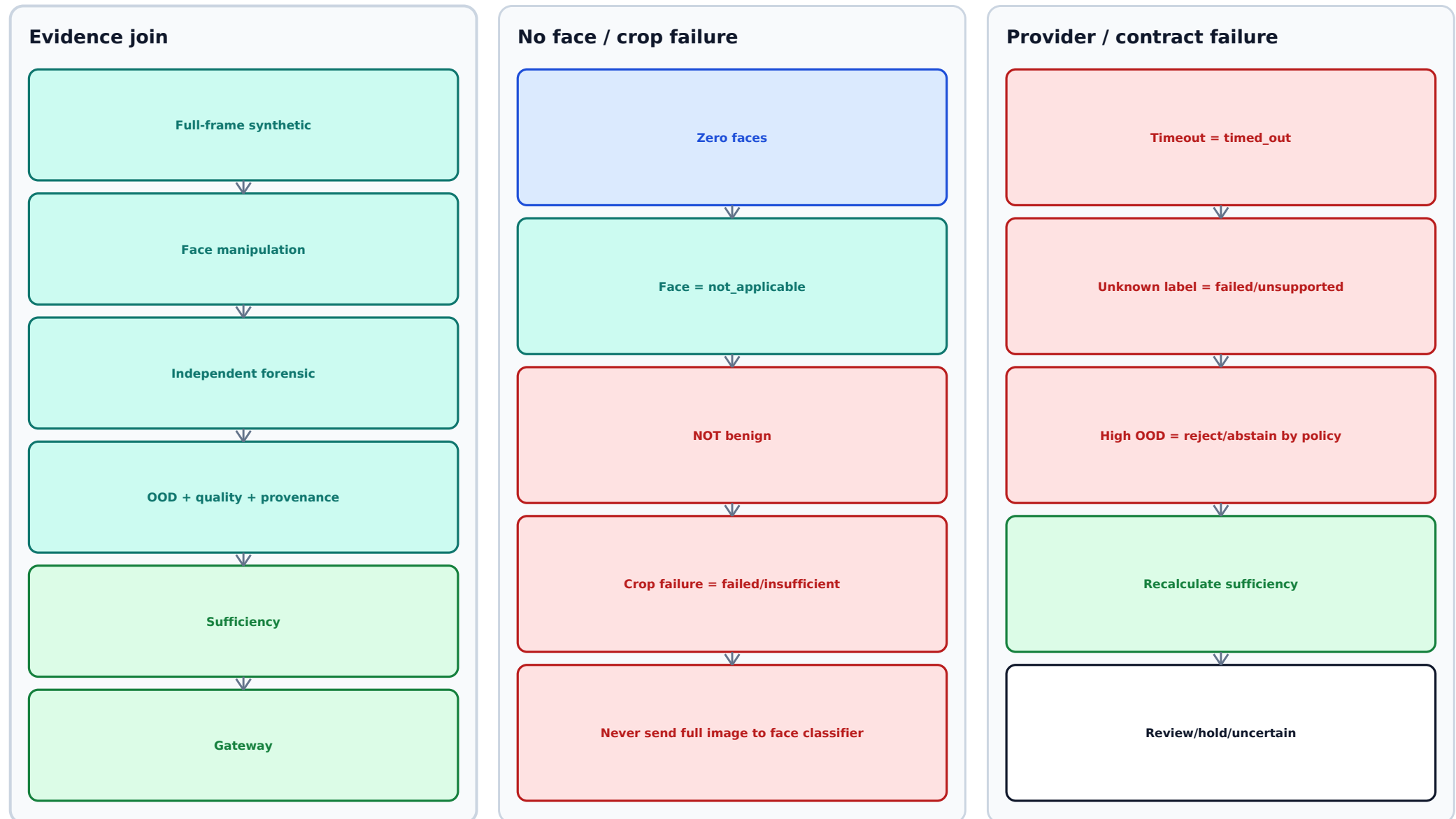
Model is not trained or qualified. No public/private URL is fabricated.

OOD / uncertainty



SigLIP2 alone is not the OOD detector.

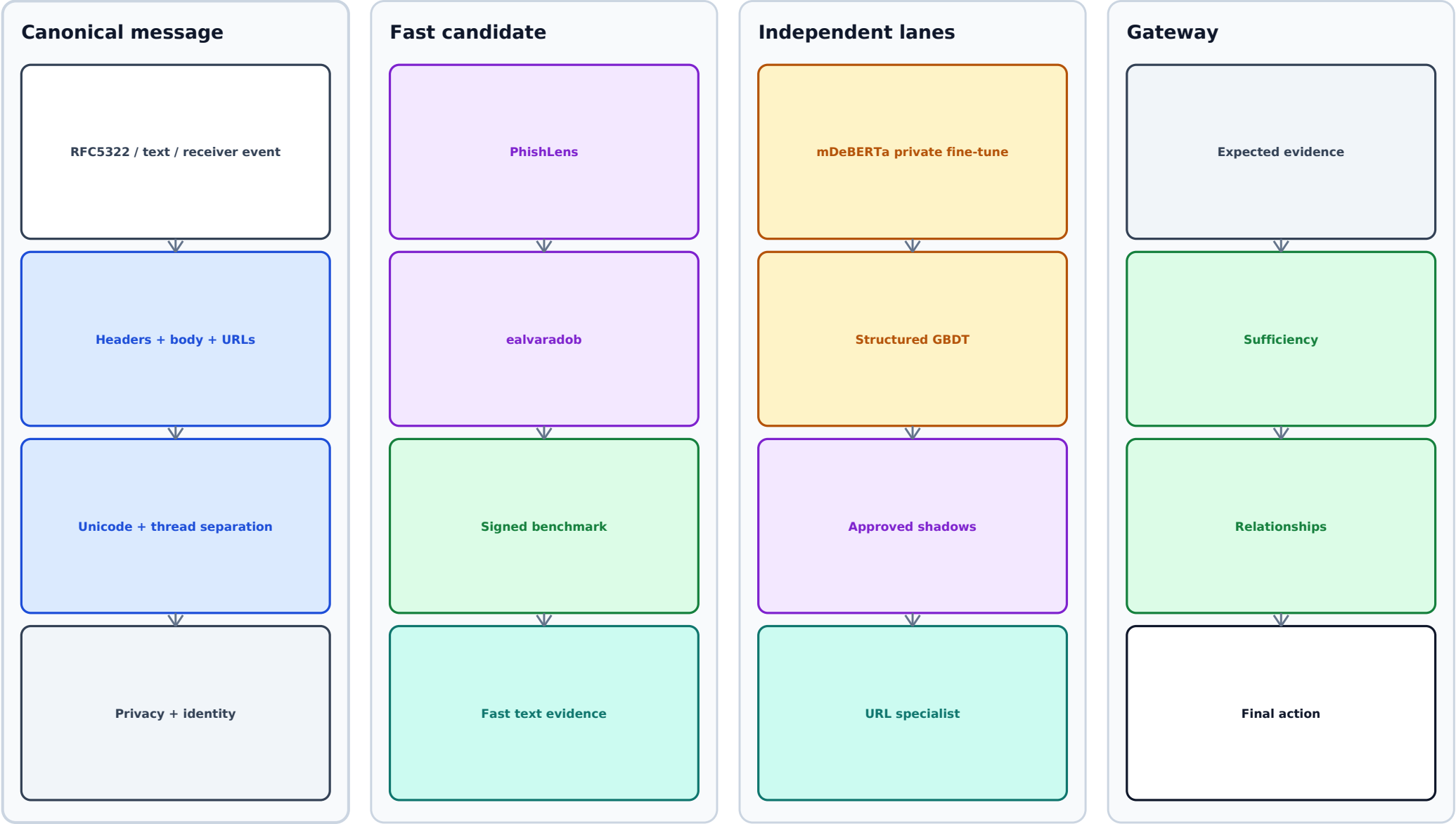
Deepfake final correlation and degraded-state flow



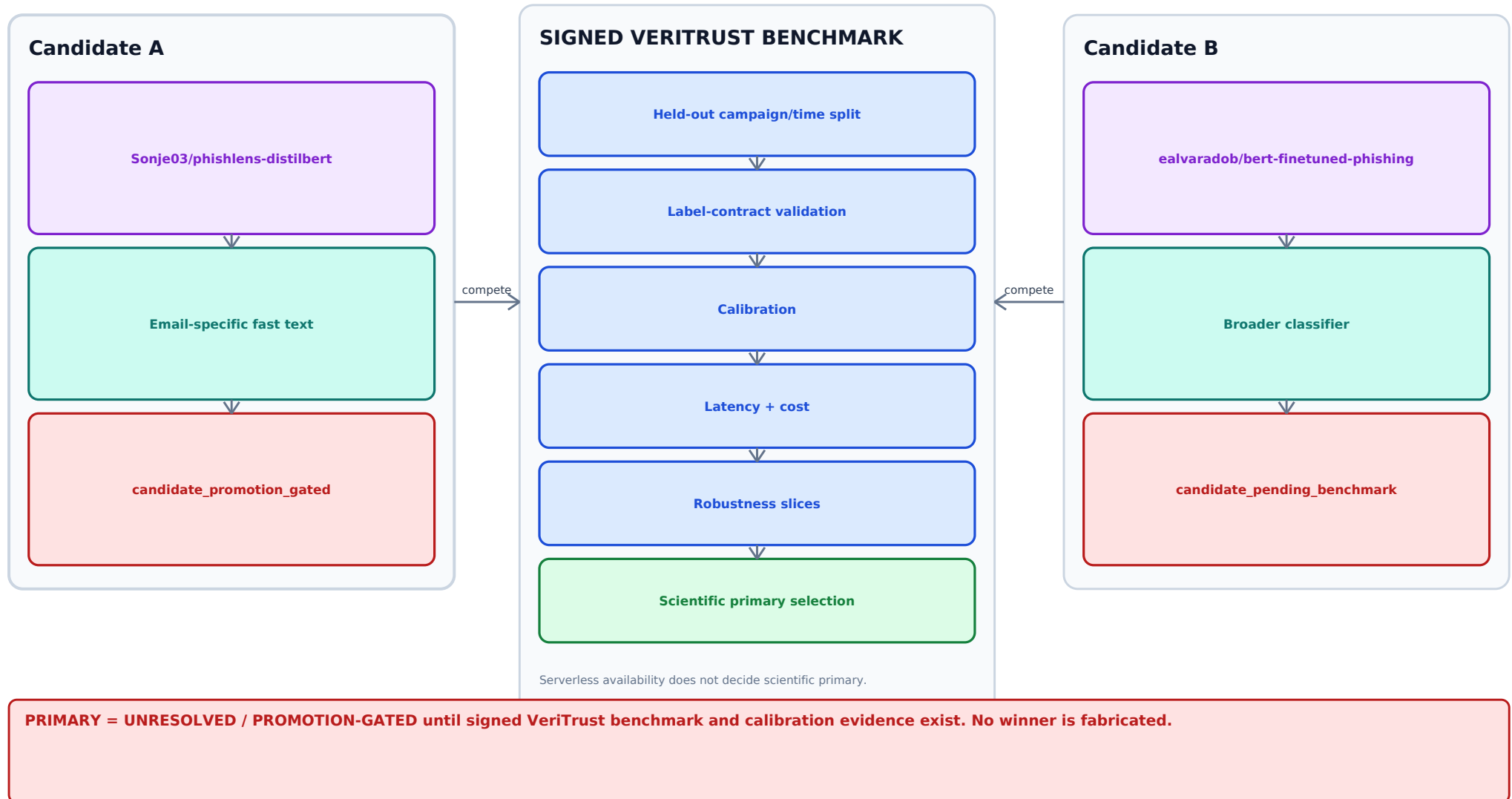
Deepfake model and evidence map

Model key	Scientific role	Repository	Revision	Deployment	Status	Authority
deepfake.face_pipeline.yunet_private_v1	face_detection_landmarks_align ment	veritrust-private/deepfake-face-pipeline-yunet-v 1	UNKNOWN	private_hf_repo_plus_hf_endpoint	requires_packaging_and_qualific ation	Evidence only; Gateway owns final action
deepfake.face_manipulation.yermandy_v1	face_manipulation	yermandy/deepfake-detection	9a6857ec642deb57373c5437b e803a199468b8c6	private_hf_repo_plus_hf_endpoint	research_candidate	Evidence only; Gateway owns final action
deepfake.synthetic.commfor384_v1	full_frame_synthesis	OwensLab/commfor-model-384	6076002bf0d9dd37537f965ee2 f06f826c333b61	hf_inference_endpoint	research_candidate	Evidence only; Gateway owns final action
deepfake.synthetic.commfor224_shadow_v1	full_frame_synthesis_shadow	OwensLab/commfor-model-224	26afc31e6b40c312c3fd42c05a 758be62446215b	hf_inference_endpoint	shadow_only	Evidence only; Gateway owns final action
deepfake.forensic.residual_private_v1	independent_low_level_forensic s	veritrust-private/deepfake-forensic-residual-v1	UNKNOWN	private_hf_repo_plus_hf_endpoint	requires_research_and_training	Evidence only; Gateway owns final action
deepfake.ood.siglip2_private_v1	deepfake_out_of_distribution	veritrust-private/deepfake-ood-siglip2	UNKNOWN	private_hf_repo_plus_hf_endpoint	requires_training	Evidence only; Gateway owns final action

Phishing: complete overview



Phishing fast-text qualification competition



Phishing multilingual, structured, and shadow lanes



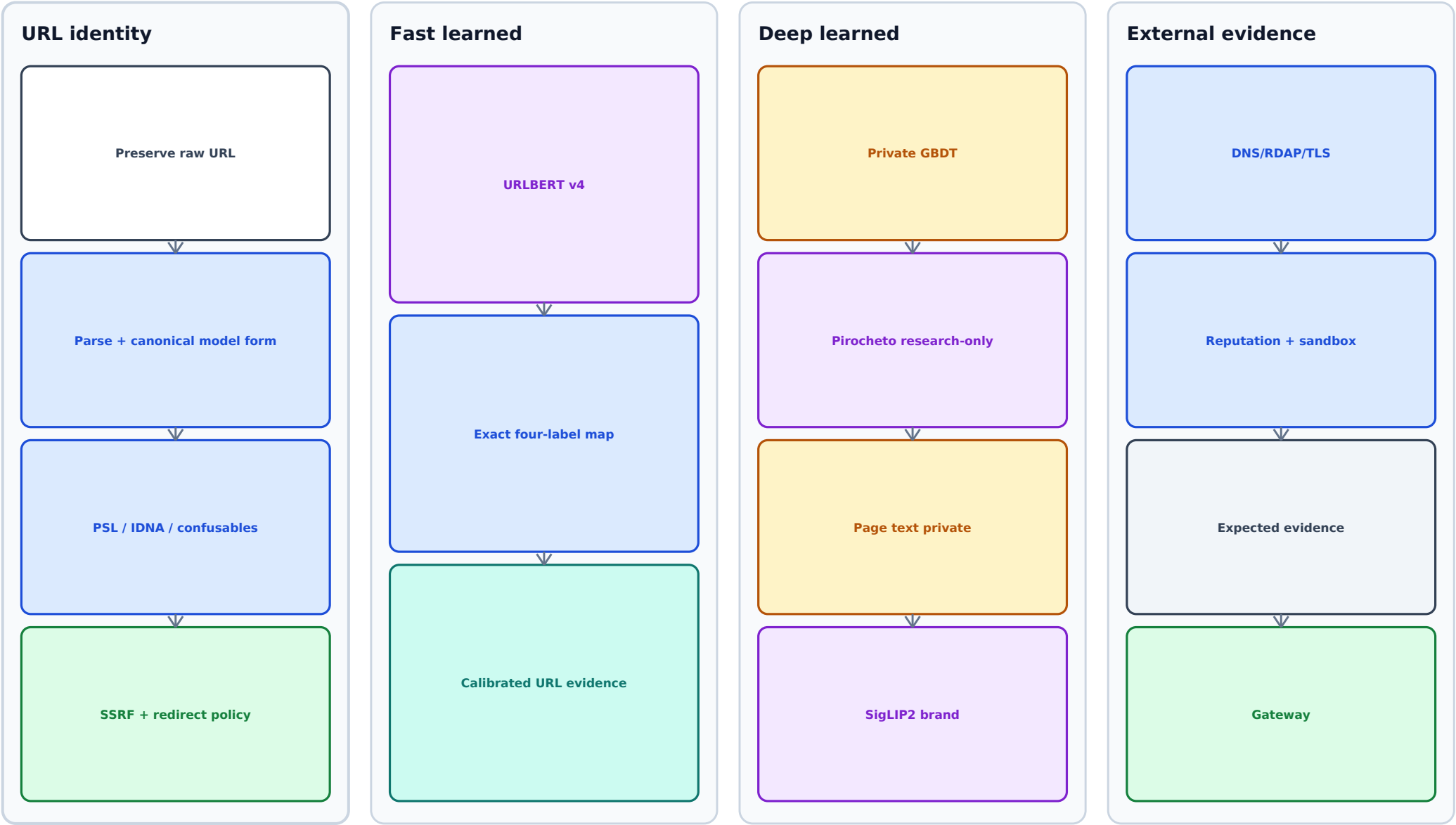
Phishing final evidence and degraded-state flow



Phishing model and evidence map

Model key	Scientific role	Repository	Revision	Deployment	Status	Authority
phishing.text.fast.phishlens_v1	phishing_text_fast_candidate	Sonje03/phishlens-distilbert	b585a2b9bbf6777b344d017f4eaff9423363a489	hf_inference_endpoint	candidate_promotion_gated	Evidence only; Gateway owns final action
phishing.text.ealvaradob_bert_v1	phishing_text_fast_candidate	ealvaradob/bert-finetuned-phishing	fa8fb73a007174c410ab7160d4e4c6e6b8d998d4	hf_serverless	candidate_pending_benchmark	Evidence only; Gateway owns final action
phishing.text.cybersectony_shadow_v1	phishing_email_text_shadow	cybersectony/phishing-email-detection-distilbert_v2.4.1	6724dec17360ab1f6450553f1d950404f66a5dde	hf_serverless	shadow_only	Evidence only; Gateway owns final action
phishing.text.mdeberta_private_v1	phishing_multilingual	veritrust-private/phishing-mdeberta-v1	UNKNOWN	private_hf_repo_plus_hf_endpoint	requires_training	Evidence only; Gateway owns final action
phishing.text.modernbert_shadow_v1	phishing_semantic_shadow	mikaelnurminen/modernbert-large-phishing	d28c16f882454115d71375d3321fbfd90e566490	hf_inference_endpoint	shadow_only	Evidence only; Gateway owns final action
phishing.features.gbd_t_private_v1	phishing_structured_features	veritrust-private/phishing-feature-gbd_t-v1	UNKNOWN	private_hf_repo_plus_hf_endpoint	requires_training	Evidence only; Gateway owns final action

Link detection: complete overview



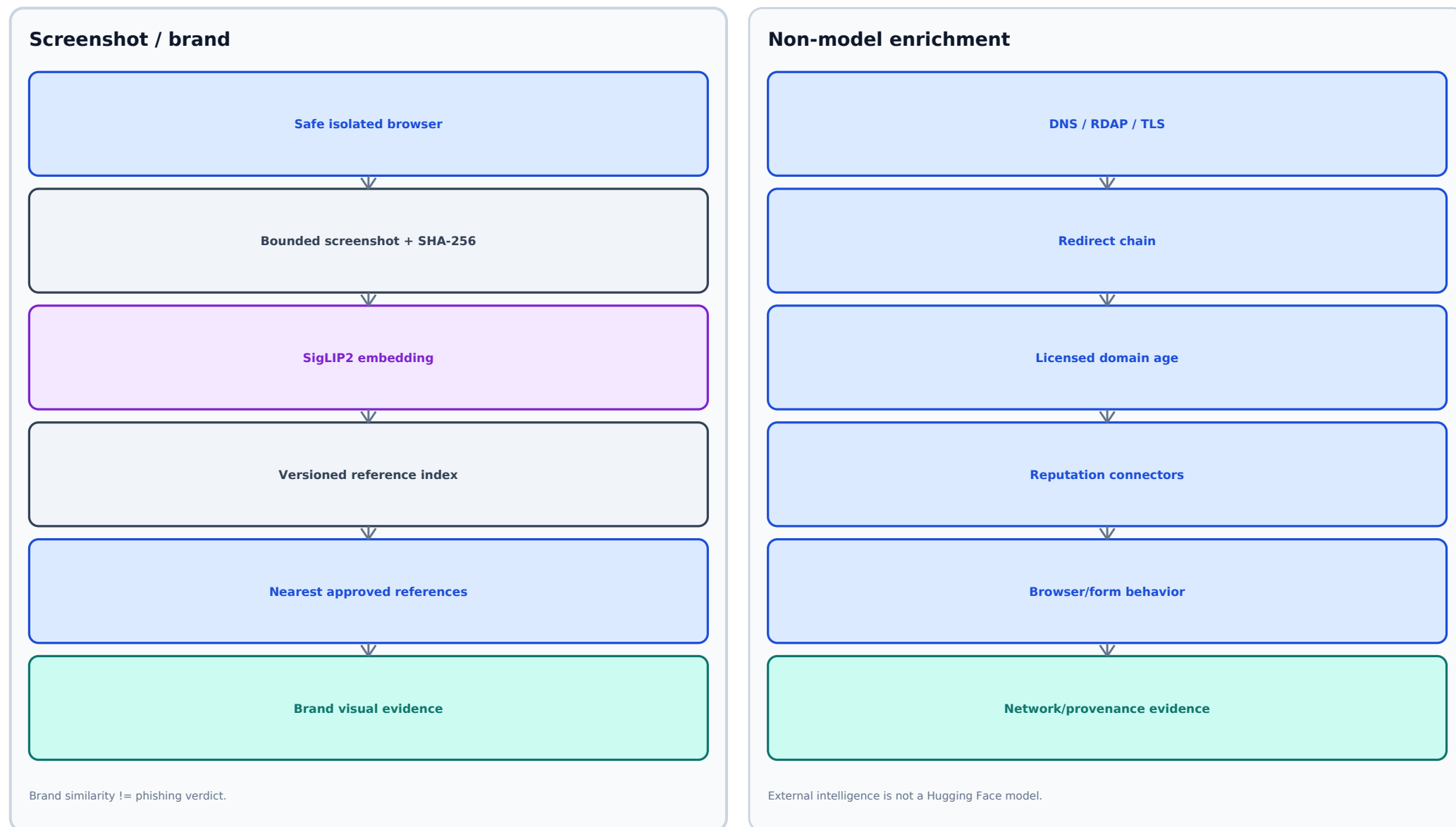
Link lane A: URLBERT four-class contract



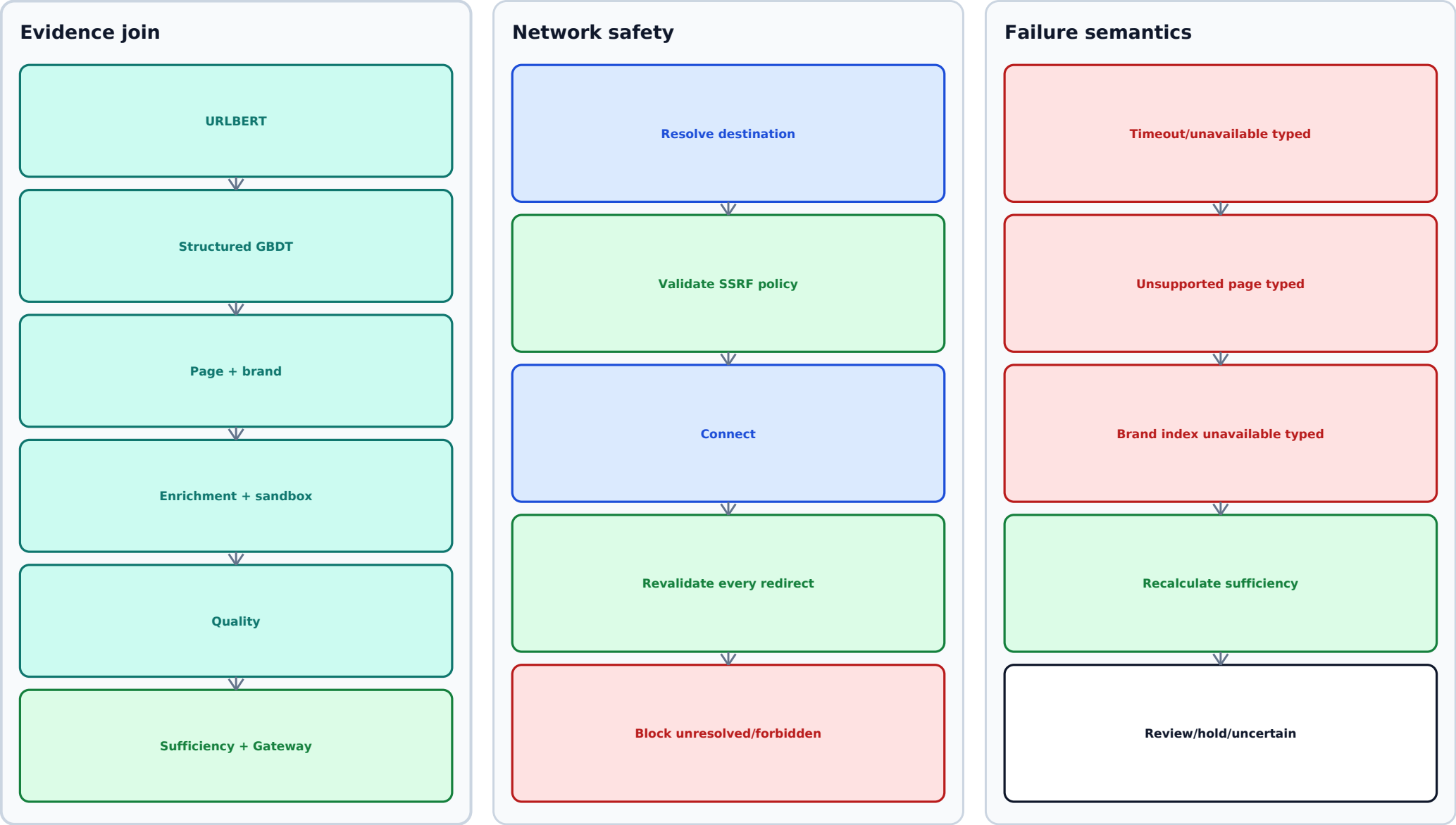
Link lanes B, C, and D: structured, shadow, page text



Link lane E + live enrichment: brand evidence



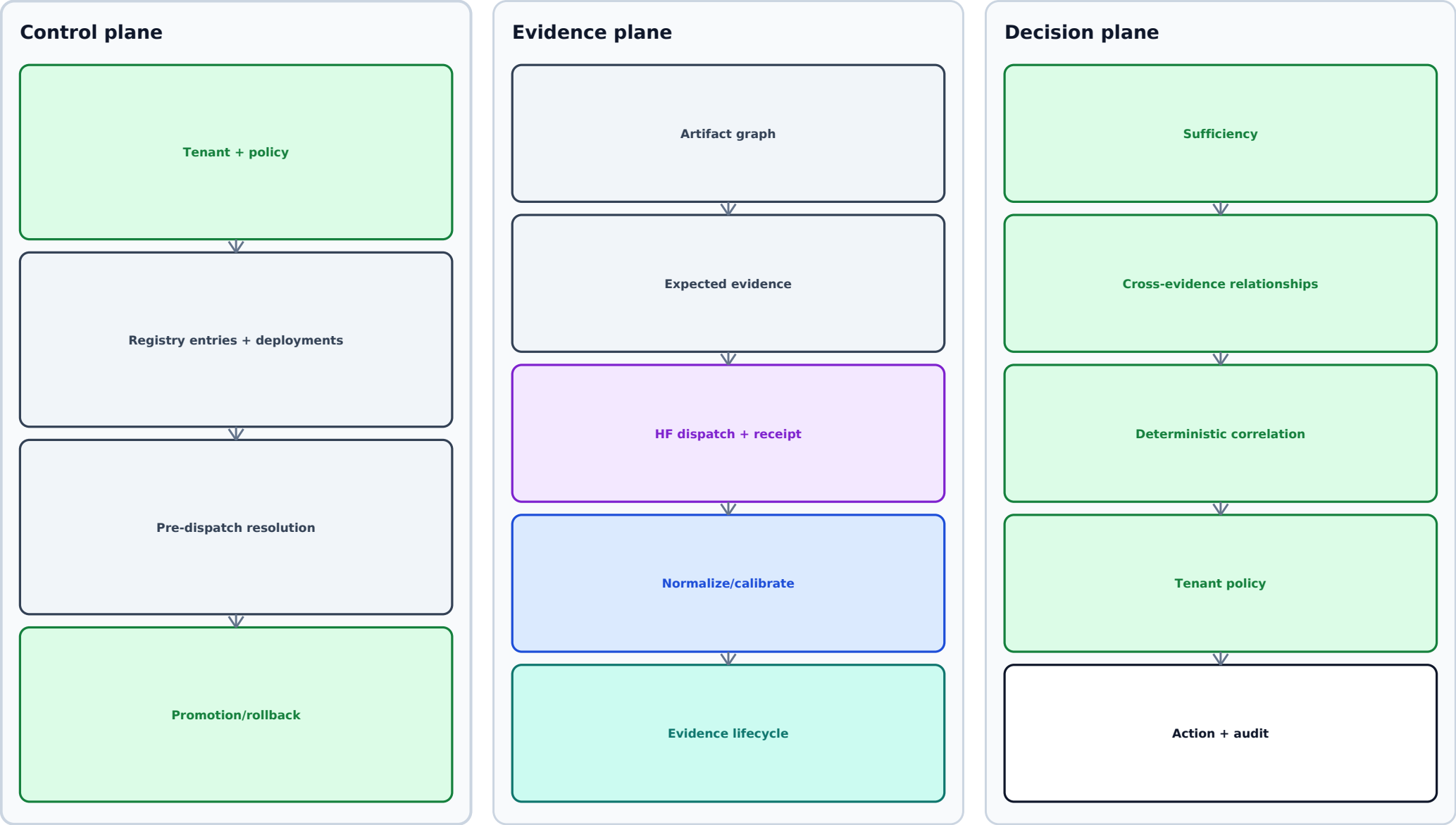
Link final evidence and degraded-state flow



Link model and evidence map

Model key	Scientific role	Repository	Revision	Deployment	Status	Authority
link.url.urlbert_v4_v1	malicious_url_primary	CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	917ded0543a630d6e82570bb01ae692f6cbb95f1	hf_inference_endpoint	candidate_pending_benchmark	Evidence only; Gateway owns final action
link.lexical.pirocheto_onnx_shadow_v1	malicious_url_lexical_shadow	pirocheto/phishing-url-detection	44f3b19f705b52532e0aadf3d0d15dd892b8a2fb	hf_inference_endpoint	research_only	Evidence only; Gateway owns final action
link.page_text.deberta_private_v1	malicious_page_text	veritrust-private/malicious-page-deberta-v1	UNKNOWN	private_hf_repo_plus_hf_endpoint	requires_training	Evidence only; Gateway owns final action
link.features.gbdt_private_v1	link_structured_features	veritrust-private/link-feature-gbdt-v1	UNKNOWN	private_hf_repo_plus_hf_endpoint	requires_training	Evidence only; Gateway owns final action
link.brand.siglip2_v1	link_screenshot_brand_embedding	google/siglip2-base-patch16-224	75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2	hf_inference_endpoint	candidate_pending_benchmark	Evidence only; Gateway owns final action

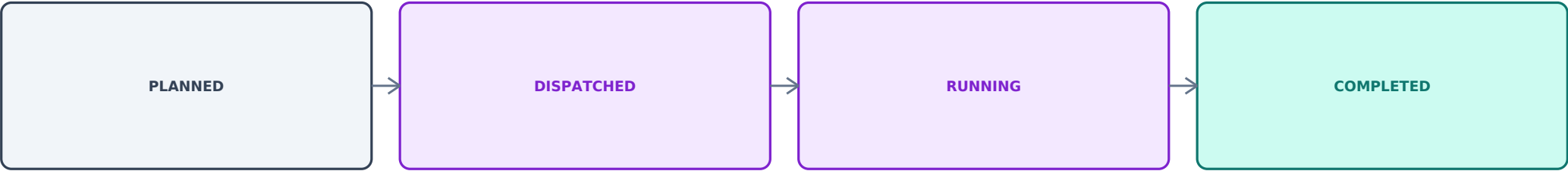
Unified Gateway: control and evidence planes



Unified Gateway: detailed implementation flow

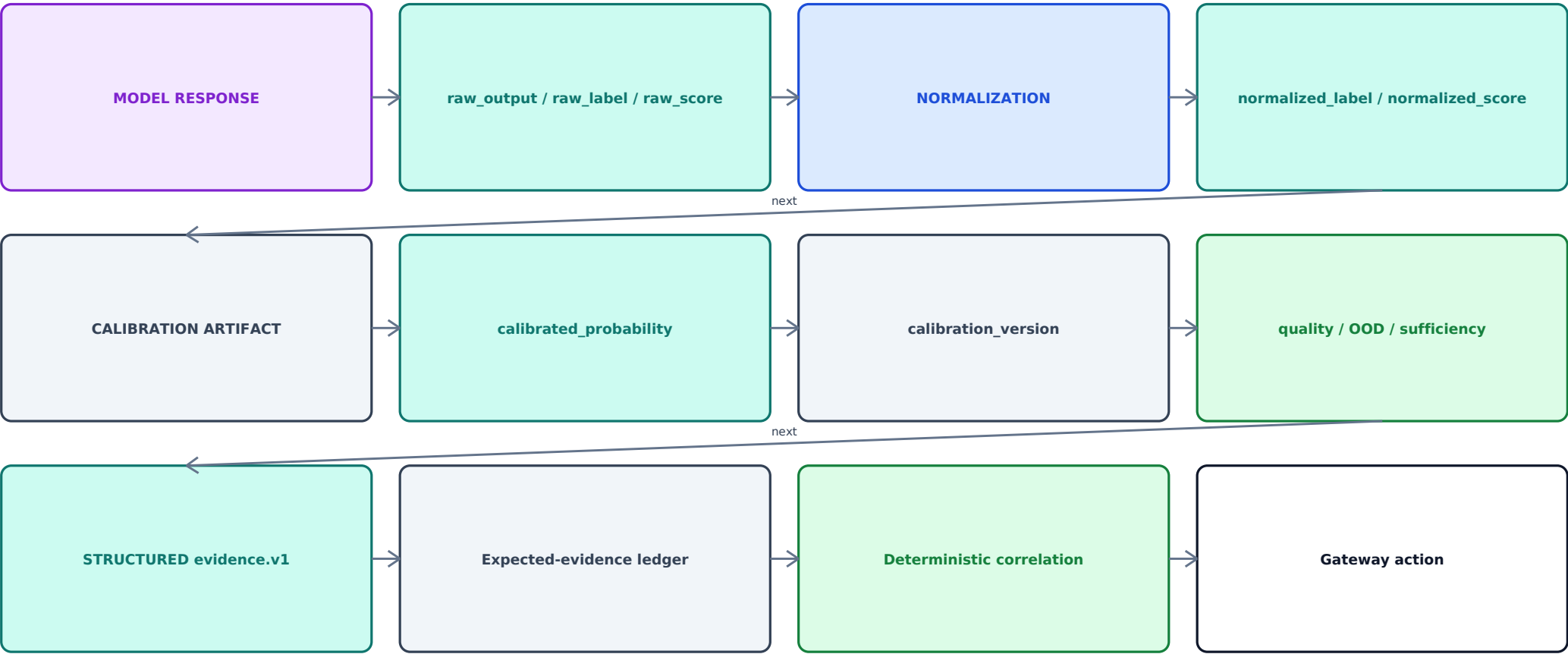


Expected-evidence lifecycle: 16 canonical states



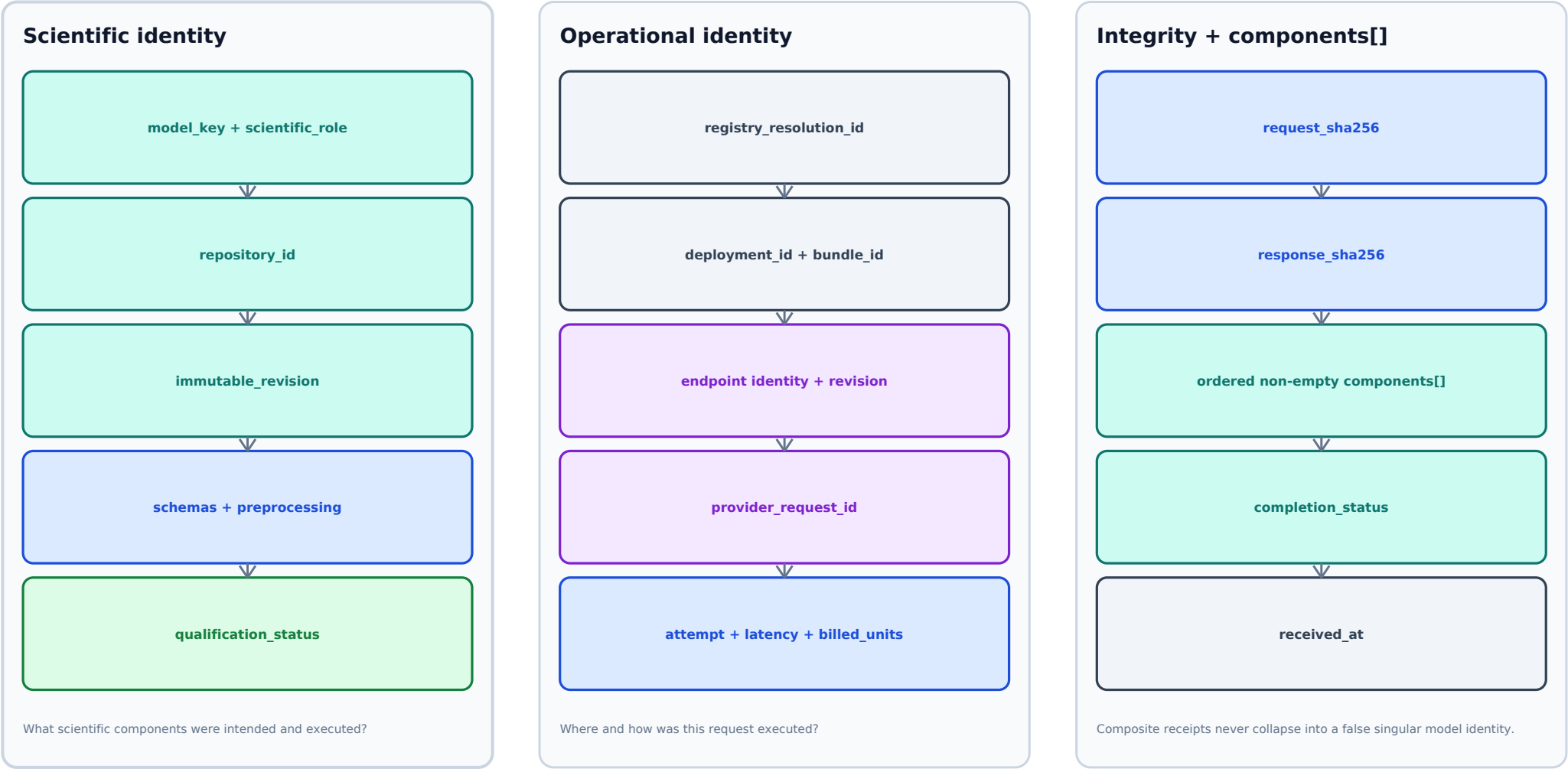
State	Terminal	Retry	Gateway may continue	Review	User visible	Meaning
planned	False	False	False	False	pending	Expected evidence has been planned but not dispatched.
dispatched	False	False	False	False	running	Immutable resolution was persisted and work was sent or queued for execution.
running	False	False	False	False	running	Execution is active.
completed	True	False	True	False	completed	Validated evidence and any required receipt are complete.
abstained	True	False	True	policy_dependent	inconclusive	Model deliberately declined a supported conclusion.
not_applicable	True	False	True	False	not_applicable	Evidence family does not apply, for example zero faces.
unavailable	True	policy_dependent	policy_dependent	policy_dependent	unavailable	Qualified deployment or dependency was unavailable after the allowed budget.
timed_out	True	policy_dependent	policy_dependent	policy_dependent	timed_out	Deadline expired before completion.
failed	True	policy_dependent	policy_dependent	policy_dependent	failed	Execution or contract validation failed.
cancelled	True	False	False	False	cancelled	Authorized cancellation stopped this expected evidence.
privacy_blocked	True	False	policy_dependent	policy_dependent	unavailable	Privacy policy prohibited dispatch or payload use.
policy_skipped	True	False	True	False	not_run	Declared policy intentionally omitted this evidence family.
insufficient	True	False	policy_dependent	policy_dependent	insufficient	Execution completed but input quality/evidence sufficiency was inadequate.
unsupported	True	False	policy_dependent	policy_dependent	unsupported	Artifact or task is outside the declared contract.
degraded	True	False	policy_dependent	policy_dependent	degraded	Only explicitly identified partial/degraded evidence is available.
budget_exceeded	True	False	policy_dependent	policy_dependent	limit_reached	Declared cost/fan-out/admission budget prevented completion.

Evidence semantic contract



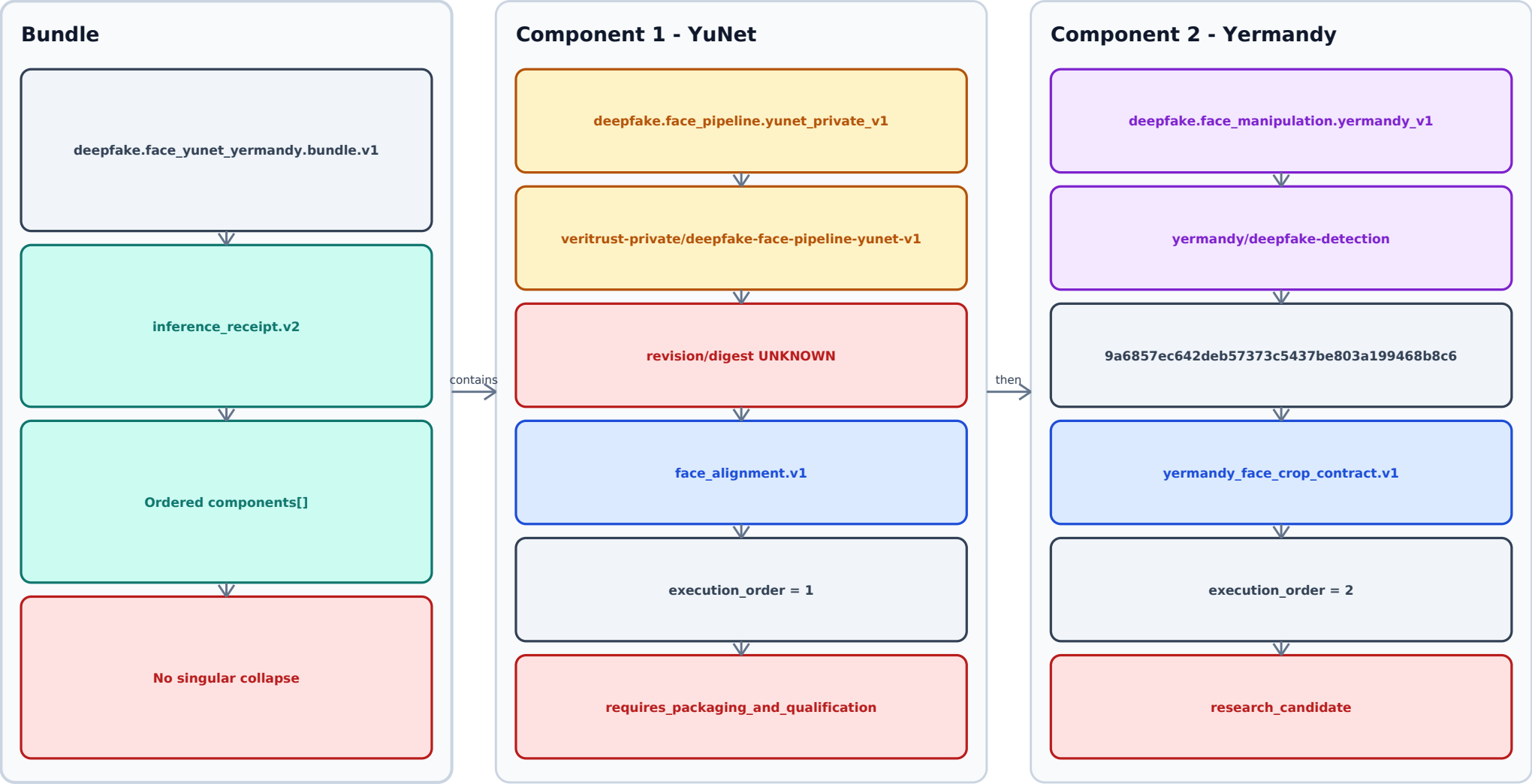
RAW SCORE != NORMALIZED SCORE != CALIBRATED PROBABILITY. Every field is separately nullable; missing calibration cannot authorize allow/block.

Inference receipt v2: provenance contract



Required receipt fields: 17 | component fields: 10 | legacy v1 accepted only through an explicit read adapter.

Composite face-pipeline receipt



Model registry map (1-3 of 18)

deepfake.face_manipulation.yermandy_v1

Domain / role: deepfake / face_manipulation
Public/private: public
Repository: yermandy/deepfake-detection
HF URL: <https://huggingface.co/yermandy/deepfake-detection>
Revision: 9a6857ec642deb57373c5437be803a199468b8c6
Input: deepfake.face_crop.v1
Output: deepfake.face_evidence.v1
Deployment: private_hf_repo_plus_hf_endpoint
Status: research_candidate
Training required: NO / qualification only
Calibration: required before promotion
HF Endpoint: YES
hf-inference: no
Exact revision gate: controlled copy; exact preprocessing/labels; cross-dataset benchmark; calibration; privacy/legal
Production qualified: NO

deepfake.face_pipeline.yunet_private_v1

Domain / role: deepfake / face_detection_landmarks_alignment
Public/private: private
Repository: veritrust-private/deepfake-face-pipeline-yunet-v1
HF URL: Private VeriTrust HF repository - not created / not qualified
Revision: UNKNOWN
Input: deepfake.full_frame.v1
Output: deepfake.face_pipeline.v1
Deployment: private_hf_repo_plus_hf_endpoint
Status: requires_packaging_and_qualification
Training required: NO / qualification only
Calibration: required before promotion
HF Endpoint: YES
hf-inference: no
Exact revision gate: OpenCV Zoo source 47534e27; artifact hash; geometry golden vectors; face-slice benchmark; privacy/security
Production qualified: NO

deepfake.forensic.residual_private_v1

Domain / role: deepfake / independent_low_level_forensics
Public/private: private
Repository: veritrust-private/deepfake-forensic-residual-v1
HF URL: Private VeriTrust HF repository - not created / not qualified
Revision: UNKNOWN
Input: deepfake.full_frame.v1
Output: deepfake.forensic_evidence.v1
Deployment: private_hf_repo_plus_hf_endpoint
Status: requires_research_and_training
Training required: YES
Calibration: required before promotion
HF Endpoint: YES
hf-inference: no
Exact revision gate: representation ablation; dataset manifest; independent-error benchmark; immutable revision; calibration
Production qualified: NO

Model registry map (4-6 of 18)

deepfake.ood.siglip2_private_v1

Domain / role:	deepfake / deepfake_out_of_distribution
Public/private:	private
Repository:	veritrust-private/deepfake-ood-siglip2
HF URL:	Private VeriTrust HF repository - not created / not qualified
Revision:	UNKNOWN
Input:	deepfake.full_frame.v1
Output:	deepfake.ood_evidence.v1
Deployment:	private_hf_repo_plus_hf_endpoint
Status:	requires_training
Training required:	YES
Calibration:	required before promotion
HF Endpoint:	YES
hf-inference:	no
Exact revision gate:	base google/siglip2-base-patch16-224@75de2d55; dataset manifest; trained head; private immutable revision; OOD benchmark
Production qualified:	NO

deepfake.synthetic.commmfor224_shadow_v1

Domain / role:	deepfake / full_frame_synthesis_shadow
Public/private:	public
Repository:	OwensLab/commfor-model-224
HF URL:	https://huggingface.co/OwensLab/commfor-model-224
Revision:	26afc31e6b40c312c3fd42c05a758be62446215b
Input:	deepfake.full_frame.v1
Output:	deepfake.synthetic_evidence.v1
Deployment:	hf_inference_endpoint
Status:	shadow_only
Training required:	NO / qualification only
Calibration:	required before promotion
HF Endpoint:	YES
hf-inference:	no
Exact revision gate:	same-family latency benchmark; calibration; never independent forensic role
Production qualified:	NO

deepfake.synthetic.commmfor384_v1

Domain / role:	deepfake / full_frame_synthesis
Public/private:	public
Repository:	OwensLab/commfor-model-384
HF URL:	https://huggingface.co/OwensLab/commfor-model-384
Revision:	6076002bf0d9dd37537f965ee2f06f826c333b61
Input:	deepfake.full_frame.v1
Output:	deepfake.synthetic_evidence.v1
Deployment:	hf_inference_endpoint
Status:	research_candidate
Training required:	NO / qualification only
Calibration:	required before promotion
HF Endpoint:	YES
hf-inference:	no
Exact revision gate:	preprocessor pin; golden outputs; generator-family benchmark; calibration
Production qualified:	NO

Model registry map (7-9 of 18)

learning.recommendation.private_v1

Domain / role:	learning / optional_advisory_learning_recommendation
Public/private:	private
Repository:	veritrust-private/learning-recommendation-v1
HF URL:	Private VeriTrust HF repository - not created / not qualified
Revision:	UNKNOWN
Input:	learning.minimized_history.v1
Output:	learning.advisory_recommendation.v1
Deployment:	private_hf_repo_plus_hf_endpoint
Status:	optional_future
Training required:	NO / qualification only
Calibration:	required before promotion
HF Endpoint:	YES
hf-inference:	no
Exact revision gate:	product approval; privacy/fairness; advisory-only contract
Production qualified:	NO

link.brand.siglip2_v1

Domain / role:	link / link_screenshot_brand_embedding
Public/private:	public
Repository:	google/siglip2-base-patch16-224
HF URL:	https://huggingface.co/google/siglip2-base-patch16-224
Revision:	75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2
Input:	link.isolated_screenshot.v1
Output:	link.brand_similarity_evidence.v1
Deployment:	hf_inference_endpoint
Status:	candidate_pending_benchmark
Training required:	NO / qualification only
Calibration:	required before promotion
HF Endpoint:	YES
hf-inference:	no
Exact revision gate:	isolated screenshot; versioned poisoning-protected brand index; retrieval benchmark; similarity calibration; supporting-only authority
Production qualified:	NO

link.features.gbdt_private_v1

Domain / role:	link / link_structured_features
Public/private:	private
Repository:	veritrust-private/link-feature-gbdt-v1
HF URL:	Private VeriTrust HF repository - not created / not qualified
Revision:	UNKNOWN
Input:	link.feature_vector.v1
Output:	link.binary_evidence.v1
Deployment:	private_hf_repo_plus_hf_endpoint
Status:	requires_training
Training required:	YES
Calibration:	required before promotion
HF Endpoint:	YES
hf-inference:	no
Exact revision gate:	feature schema; dataset; training; artifact hash; SBOM; calibration
Production qualified:	NO

Model registry map (10-12 of 18)

link.lexical.pirocheto_onnx_shadow_v1 Domain / role: Public/private: Repository: HF URL: Revision: Input: Output: Deployment: Status: Training required: Calibration: HF Endpoint: hf-inference: Exact revision gate: Production qualified: link / malicious_url_lexical_shadow public pirocheto/phishing-url-detection https://huggingface.co/pirocheto/phishing-url-detection 44f3b19f705b52532e0aadf3d0d15dd892b8a2fb link.lexical_features.v1 link.binary_evidence.v1 hf_inference_endpoint research_only NO / qualification only required before promotion YES no safe artifact selection; I/O semantics; legal review; benchmark; calibration NO	link.page_text.deberta_private_v1 Domain / role: Public/private: Repository: HF URL: Revision: Input: Output: Deployment: Status: Training required: Calibration: HF Endpoint: hf-inference: Exact revision gate: Production qualified: link / malicious_page_text private veritrust-private/malicious-page-deberta-v1 Private VeriTrust HF repository - not created / not qualified UNKNOWN link.sanitized_page_text.v1 link.binary_evidence.v1 private_hf_repo_plus_hf_endpoint requires_training YES required before promotion YES no base microsoft/deberta-v3-small@a36c7390; safe extraction; training; private immutable revision; benchmark; calibration NO	link.url.urlbert_v4_v1 Domain / role: Public/private: Repository: HF URL: Revision: Input: Output: Deployment: Status: Training required: Calibration: HF Endpoint: hf-inference: Exact revision gate: Production qualified: link / malicious_url_primary public CrabInHoney/urlbert-tiny-v4-malicious-url-classifier https://huggingface.co/CrabInHoney/urlbert-tiny-v4-malicious-url-classifier 917ded0543a630d6e82570bb01ae692f6cbb95f1 link.canonical_model_string.v1 link.four_class_evidence.v1 hf_inference_endpoint candidate_pending_benchmark NO / qualification only required before promotion YES no fix LABEL_1/LABEL_3 inversion; tokenizer/normalization parity; benchmark; calibration NO
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Model registry map (13-15 of 18)

phishing.features.gbdt_private_v1

Domain / role: phishing / phishing_structured_features
Public/private: private
Repository: veritrust-private/phishing-feature-gbdt-v1
HF URL: Private VeriTrust HF repository - not created / not qualified
Revision: UNKNOWN
Input: phishing.feature_vector.v1
Output: phishing.binary_evidence.v1
Deployment: private_hf_repo_plus_hf_endpoint
Status: requires_training
Training required: YES
Calibration: required before promotion
HF Endpoint: YES
hf-inference: no
Exact revision gate: feature schema; dataset; training; artifact hash; SBOM; calibration
Production qualified: NO

phishing.text.cybersectony_shadow_v1

Domain / role: phishing / phishing_email_text_shadow
Public/private: public
Repository: cybersectony/phishing-email-detection-distilbert_v2.4.1
HF URL: https://huggingface.co/cybersectony/phishing-email-detection-distilbert_v2.4.1
Revision: 6724dec17360ab1f6450553f1d950404f66a5dde
Input: phishing.plain_text.v1
Output: phishing.four_label_raw.v1
Deployment: hf_serverless
Status: shadow_only
Training required: NO / qualification only
Calibration: required before promotion
HF Endpoint: conditional
hf-inference: no until label semantics proven
Exact revision gate: authoritative four-label mapping; task-boundary review; golden vectors; calibration
Production qualified: NO

phishing.text.ealvaradob_bert_v1

Domain / role: phishing / phishing_text_fast_candidate
Public/private: public
Repository: ealvaradob/bert-finetuned-phishing
HF URL: <https://huggingface.co/ealvaradob/bert-finetuned-phishing>
Revision: fa8fb73a007174c410ab7160d4e4c6e6b8d998d4
Input: phishing.plain_text.v1
Output: phishing.binary_evidence.v1
Deployment: hf_serverless
Status: candidate_pending_benchmark
Training required: NO / qualification only
Calibration: required before promotion
HF Endpoint: conditional
hf-inference: yes after scientific gates
Exact revision gate: head-to-head with PhishLens; golden request; tokenizer/truncation; temporal/domain benchmark; calibration; quota/privacy
Production qualified: NO

Model registry map (16-18 of 18)

phishing.text.fast.phishlens_v1

Domain / role: phishing / phishing_text_fast_candidate
Public/private: public
Repository: Sonje03/phishlens-distilbert
HF URL: <https://huggingface.co/Sonje03/phishlens-distilbert>
Revision: b585a2b9bbf6777b344d017f4eaff9423363a489
Input: phishing.email_text.v1
Output: phishing.binary_evidence.v1
Deployment: hf_inference_endpoint
Status: candidate_promotion_gated
Training required: NO / qualification only
Calibration: required before promotion
HF Endpoint: YES
hf-inference: no
Exact revision gate: head-to-head with ealvaradob; tokenizer/truncation; temporal/domain benchmark; calibration; latency/privacy/license
Production qualified: NO

phishing.text.mdeberta_private_v1

Domain / role: phishing / phishing_multilingual
Public/private: private
Repository: veritrust-private/phishing-mdeberta-v1
HF URL: Private VeriTrust HF repository - not created / not qualified
Revision: UNKNOWN
Input: phishing.localized_text.v1
Output: phishing.binary_evidence.v1
Deployment: private_hf_repo_plus_hf_endpoint
Status: requires_training
Training required: YES
Calibration: required before promotion
HF Endpoint: YES
hf-inference: no
Exact revision gate: base microsoft/mdeberta-v3-base@a0484667; train classifier; language slices; immutable private revision; calibration
Production qualified: NO

phishing.text.modernbert_shadow_v1

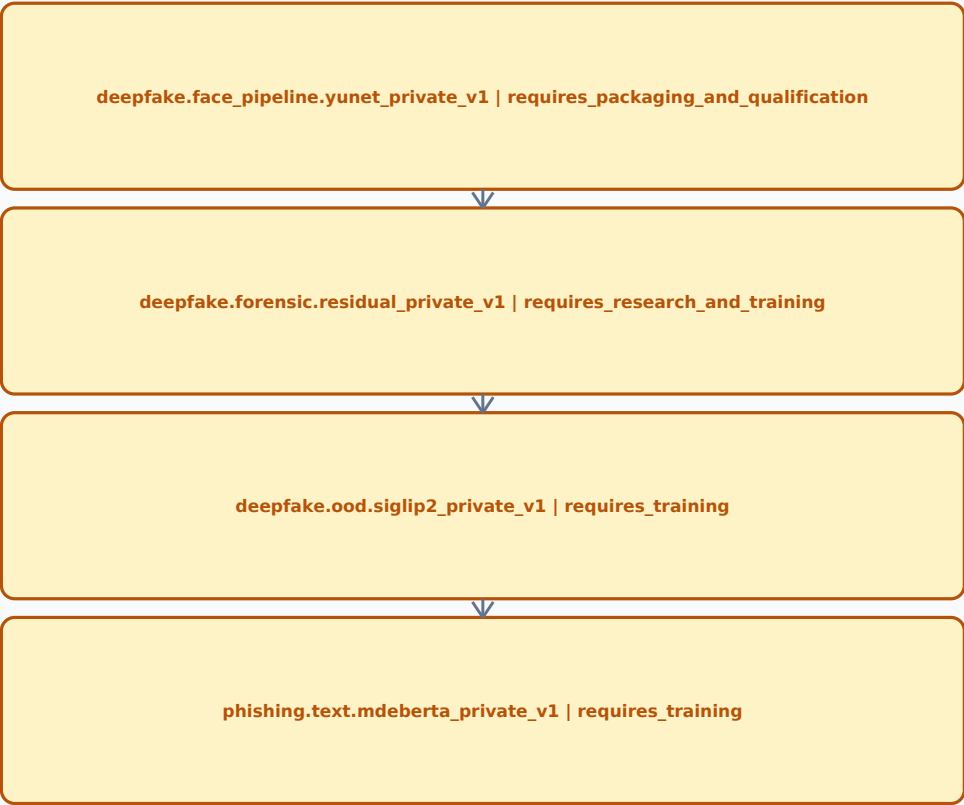
Domain / role: phishing / phishing_semantic_shadow
Public/private: public
Repository: mikaelnurminen/modernbert-large-phishing
HF URL: <https://huggingface.co/mikaelnurminen/modernbert-large-phishing>
Revision: d28c16f882454115d71375d3321fbfd90e566490
Input: phishing.plain_text.v1
Output: phishing.binary_evidence.v1
Deployment: hf_inference_endpoint
Status: shadow_only
Training required: NO / qualification only
Calibration: required before promotion
HF Endpoint: YES
hf-inference: no
Exact revision gate: endpoint deploy; independence analysis; benchmark; calibration; cost
Production qualified: NO

Public Hugging Face model quick reference

Domain	Official HF repository	Clickable URL	Immutable research revision	Architectural role	Status
deepfake	yermandy/deepfake-detection	https://huggingface.co/yermandy/deepfake-detection	9a6857ec642deb57373c5437be803a199468b8c6	face_manipulation	research_candidate
deepfake	OwensLab/commfor-model-224	https://huggingface.co/OwensLab/commfor-model-224	26afc31e6b40c312c3fd42c05a758be62446215b	full_frame_synthesis_shadow	shadow_only
deepfake	OwensLab/commfor-model-384	https://huggingface.co/OwensLab/commfor-model-384	6076002bf0d9dd37537f965ee2f06f826c333b61	full_frame_synthesis	research_candidate
link	google/siglip2-base-patch16-224	https://huggingface.co/google/siglip2-base-patch16-224	75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2	link_screenshot_brand_embedding	candidate_pending_benchmark
link	pirocheto/phishing-url-detection	https://huggingface.co/pirocheto/phishing-url-detection	44f3b19f705b52532e0aadf3d0d15dd892b8a2fb	malicious_url_lexical_shadow	research_only
link	CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	https://huggingface.co/CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	917ded0543a630d6e82570bb01ae692f6cbb95f1	malicious_url_primary	candidate_pending_benchmark
phishing	cybersectony/phishing-email-detection-distilbert_v2.4.1	https://huggingface.co/cybersectony/phishing-email-detection-distilbert_v2.4.1	6724dec17360ab1f6450553f1d950404f66a5dde	phishing_email_text_shadow	shadow_only
phishing	ealvaradob/bert-finetuned-phishing	https://huggingface.co/ealvaradob/bert-finetuned-phishing	fa8fb73a007174c410ab7160d4e4c6e6b8d998d4	phishing_text_fast_candidate	candidate_pending_benchmark
phishing	Sonje03/phishlens-distilbert	https://huggingface.co/Sonje03/phishlens-distilbert	b585a2b9bbf6777b344d017f4eaff9423363a489	phishing_text_fast_candidate	candidate_promotion_gated
phishing	mikaelnurminen/modernbert-large-phishing	https://huggingface.co/mikaelnurminen/modernbert-large-phishing	d28c16f882454115d71375d3321bfd90e566490	phishing_semantic_shadow	shadow_only
phishing	microsoft/mdeberta-v3-base	https://huggingface.co/microsoft/mdeberta-v3-base	a0484667b22365f84929a935b5e50a51f71f159d	base for private multilingual phishing classifier	base_checkpoint_not_classifier
link	microsoft/deberta-v3-small	https://huggingface.co/microsoft/deberta-v3-small	a36c739020e01763fe789b4b85e2df55d6180012	base for private page-text classifier	base_checkpoint_not_classifier

Private VeriTrust learned roles: training roadmap

Private roles 1-4



DATA -> TRAIN -> HELD-OUT EVAL -> CALIBRATE -> PRIVATE HF REPO -> IMMUTABLE REVISION -> HF ENDPOINT -> SHADOW -> CANARY -> QUALIFY

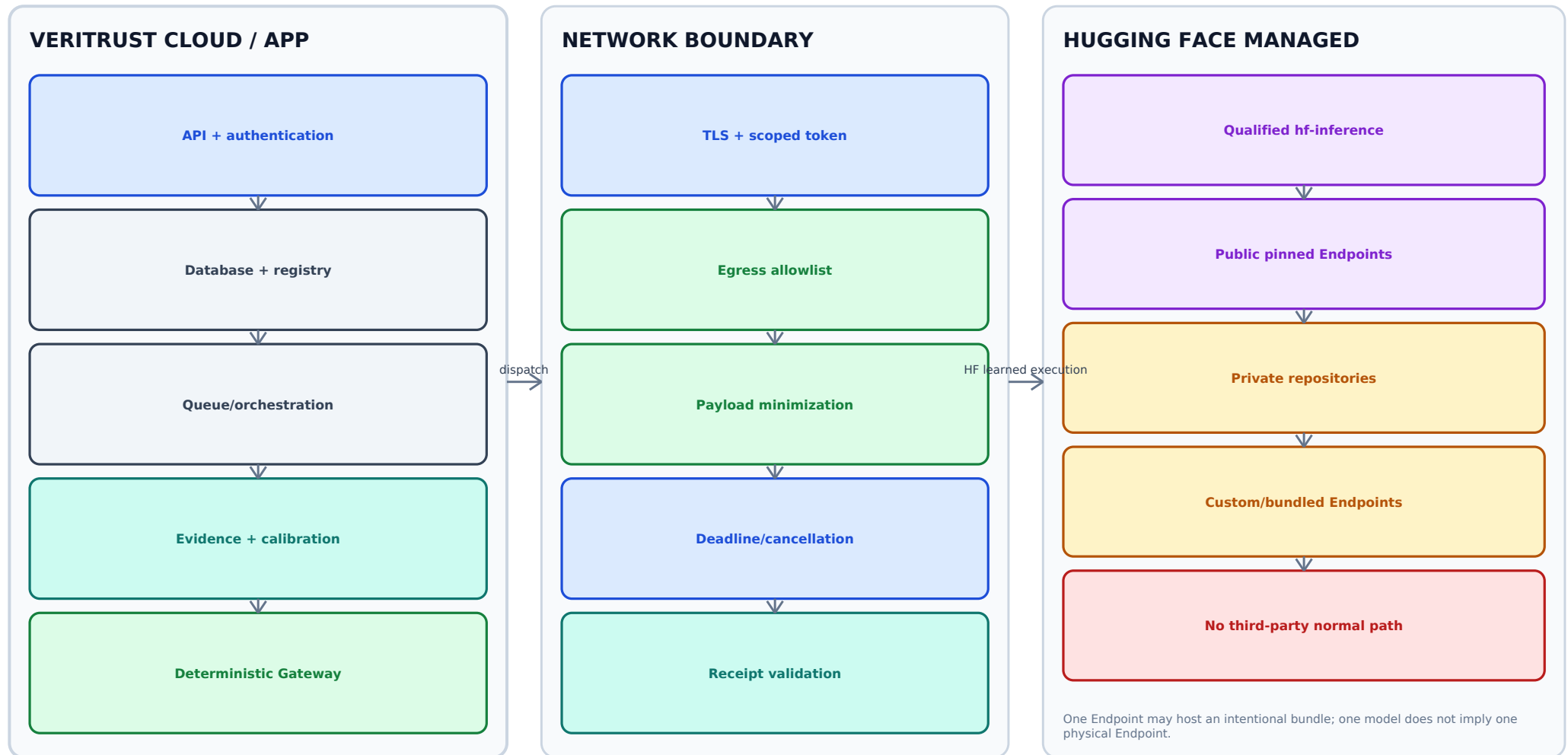
Private roles 5-8



DATA -> TRAIN -> HELD-OUT EVAL -> CALIBRATE -> PRIVATE HF REPO -> IMMUTABLE REVISION -> HF ENDPOINT -> SHADOW -> CANARY -> QUALIFY

None of these private learned roles is represented as already trained or production-qualified. UNKNOWN revision/digest remains an explicit promotion blocker.

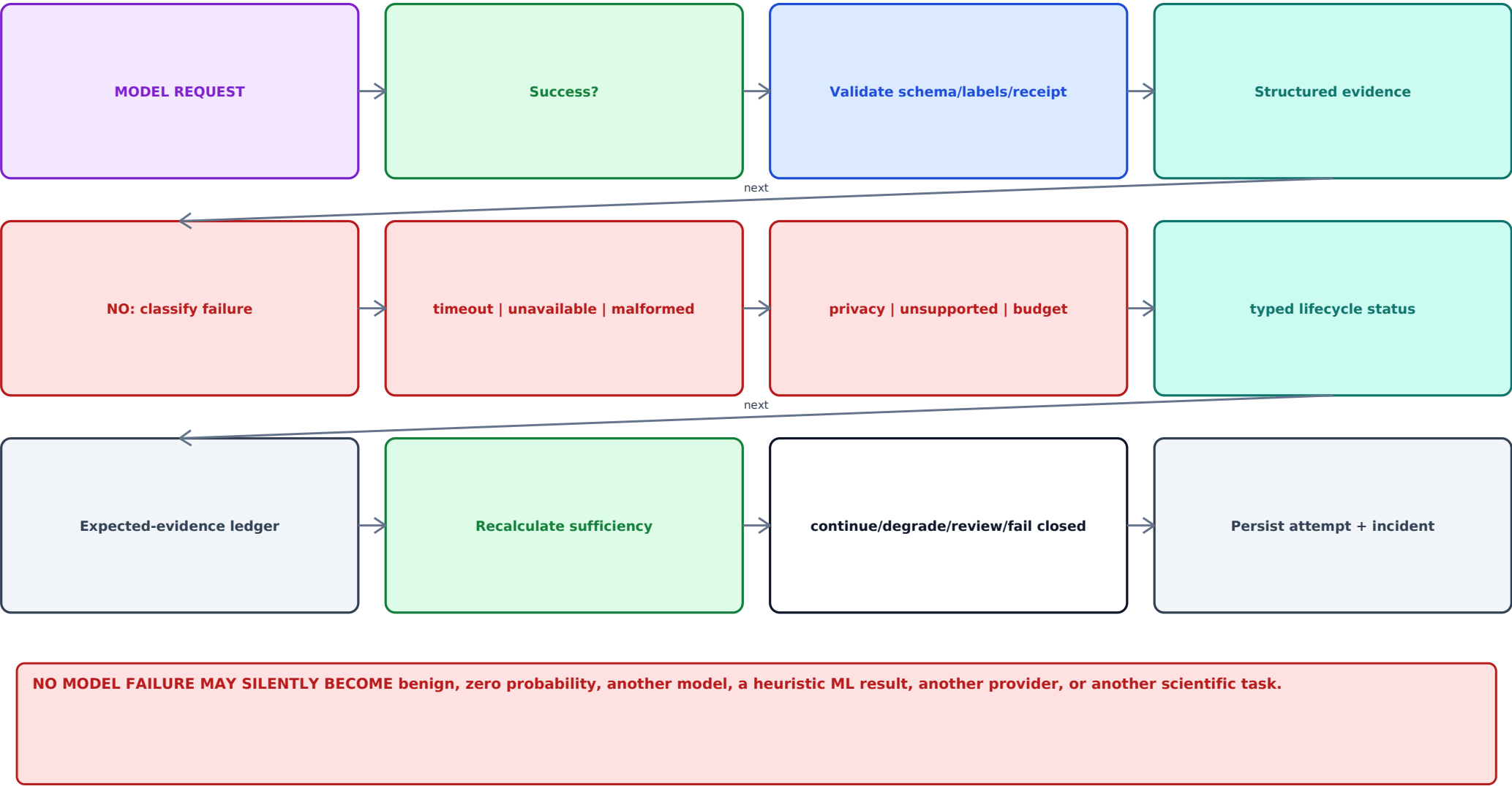
Deployment topology



Sensitive-data privacy flows



Failure behavior master flow



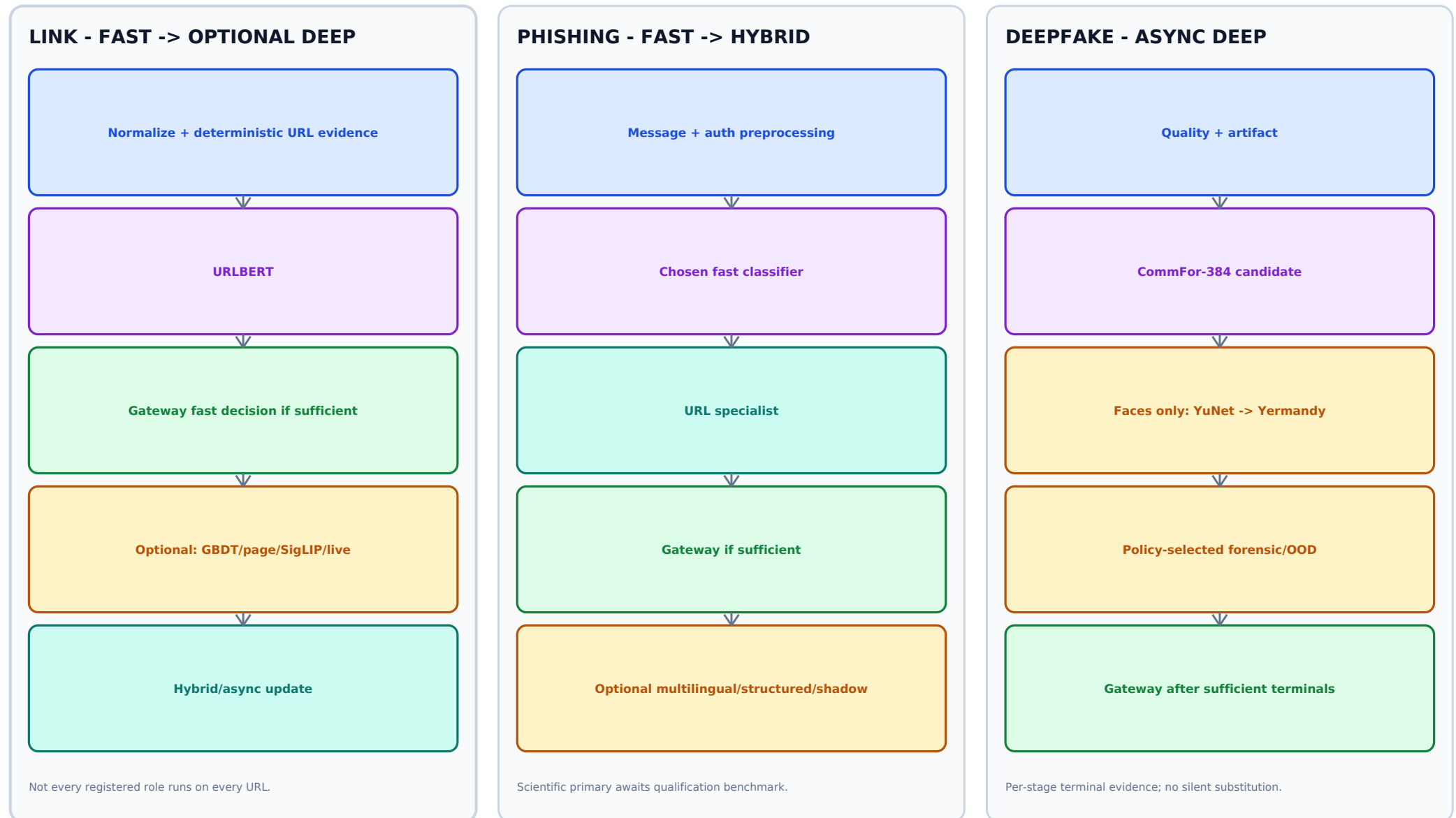
Illustrative decision examples - no thresholds



Implementation sequence



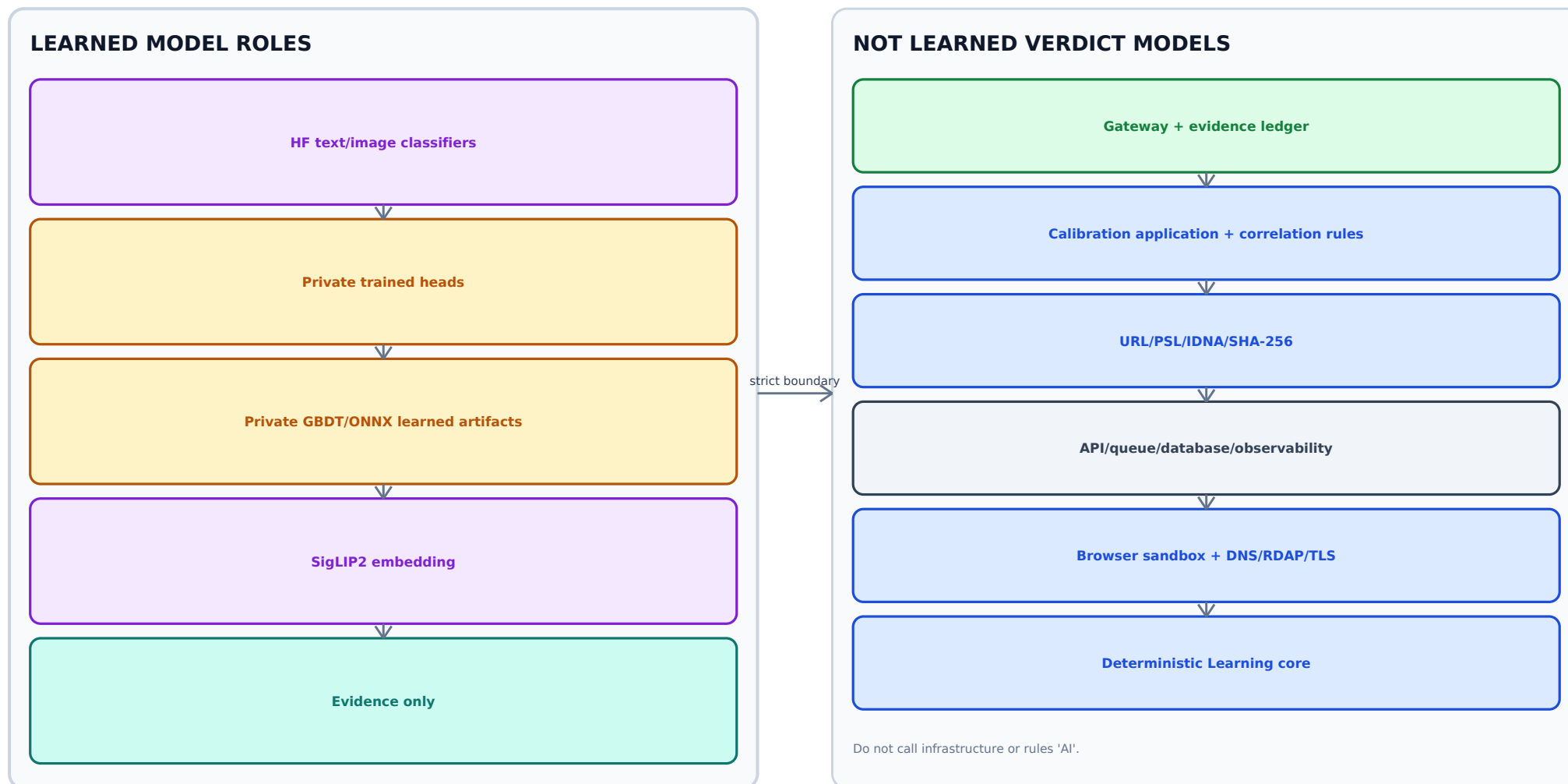
What VeriTrust actually runs per request



What each model does - simple language

Model / role	Plain-language purpose
CommFor-384	Looks at the complete image for patterns associated with synthetic/generated imagery.
CommFor-224	A smaller same-family shadow used for latency/error comparison, not independent forensic evidence.
Yermandy	Examines properly detected and aligned faces for manipulation evidence.
Private forensic model	Looks at residual, SRM-style and frequency evidence unlike normal semantic image understanding.
PhishLens	Examines email text for phishing patterns as one fast candidate.
ealvaradob BERT	A broader phishing classifier competing in the same signed qualification benchmark.
mDeBERTa private head	A future trained multilingual phishing classifier; the base alone is not a detector.
URLBERT v4	Examines URL structure/tokens and returns one of four exact URL classes.
Pirocheto	A reviewed ONNX research-only lexical URL comparison lane.
SigLIP2	Creates embeddings for approved brand-reference similarity and as a base for a private OOD head.
Private page-text model	A future DeBERTa-based classifier using sanitized visible page text.

What is not a model



Final readiness map

Readiness class	Authoritative meaning
READY TO IMPLEMENT	Canonical contracts; evidence.v1; receipt.v2; registry-before-dispatch; URLBERT label adapter architecture; deterministic Gateway/Learning core
NEEDS BENCHMARK	PhishLens vs ealvaradob primary; CommFor-384; URLBERT; SigLIP2 brand; Yermandy; all candidate calibrations
NEEDS TRAINING	Private forensic residual/SRM/DCT; private SigLIP2 OOD head; multilingual mDeBERTa; phishing GBDT; URL GBDT; page-text DeBERTa
EXTERNAL OPERATIONAL QUALIFICATION	Endpoint IDs/regions/quotas; measured SLO/latency/throughput/cost; live database drift; legal/DPA/retention; capacity; incident drills

Architecture readiness is not scientific or operational qualification. No endpoint, benchmark, calibration, cost, SLO, quota, retention period or legal approval is fabricated.

Official Hugging Face sources

Repository	Official HF URL	Research revision	Frozen role	Status
yermandy/deepfake-detection	https://huggingface.co/yermandy/deepfake-detection	9a6857ec642deb57373c5437be803a199468b8c6	face_manipulation	research_candidate
OwensLab/commfor-model-224	https://huggingface.co/OwensLab/commfor-model-224	26afc31e6b40c312c3fd42c05a758be62446215b	full_frame_synthesis_shadow	shadow_only
OwensLab/commfor-model-384	https://huggingface.co/OwensLab/commfor-model-384	6076002bf0d9dd37537f965ee2f06f826c333b61	full_frame_synthesis	research_candidate
google/siglip2-base-patch16-224	https://huggingface.co/google/siglip2-base-patch16-224	75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2	link_screenshot_brand_embedding	candidate_pending_benchmark
pirocheto/phishing-url-detection	https://huggingface.co/pirocheto/phishing-url-detection	44f3b19f705b52532e0aadf3d0d15dd892b8a2fb	malicious_url_lexical_shadow	research_only
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	https://huggingface.co/CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	917ded0543a630d6e82570bb01ae692f6cbb95f1	malicious_url_primary	candidate_pending_benchmark
cybersectony/phishing-email-detection-distilbert_v2.4.1	https://huggingface.co/cybersectony/phishing-email-detection-distilbert_v2.4.1	6724dec17360ab1f6450553f1d950404f66a5dde	phishing_email_text_shadow	shadow_only
ealvaradob/bert-finetuned-phishing	https://huggingface.co/ealvaradob/bert-finetuned-phishing	fa8fb73a007174c410ab7160d4e4c6e6b8d998d4	phishing_text_fast_candidate	candidate_pending_benchmark
Sonje03/phishlens-distilbert	https://huggingface.co/Sonje03/phishlens-distilbert	b585a2b9bbf6777b344d017f4eaff9423363a489	phishing_text_fast_candidate	candidate_promotion_gated
mikaelnurminen/modernbert-large-phishing	https://huggingface.co/mikaelnurminen/modernbert-large-phishing	d28c16f882454115d71375d3321fbfd90e566490	phishing_semantic_shadow	shadow_only
microsoft/mdeberta-v3-base	https://huggingface.co/microsoft/mdeberta-v3-base	a0484667b22365f84929a935b5e50a51f71f159d	base for private multilingual phishing classifier	base_checkpoint_not_classifier
microsoft/deberta-v3-small	https://huggingface.co/microsoft/deberta-v3-small	a36c739020e01763fe789b4b85e2df55d6180012	base for private page-text classifier	base_checkpoint_not_classifier

Canonical package sources and terminology

Canonical package source	Use in this document
Architecture_Source_of_Truth.yaml	Canonical invariants, roles, schemas, lifecycle, receipts, rollout
Model_Registry_Proposal.json	18 executable model roles and immutable research identities
HuggingFace_Deployment_Matrix.csv	Deployment classes, provider mapping, handlers, gates
HF_Model_Decision_Ledger.csv	15 investigated repository dispositions
Database_Target_Contract.md	Control-plane and evidence persistence contract
Database_Migration_Specification.md	M0-M6 expand/dual-write/backfill/switch/enforce/contract
Migration_Roadmap.md	P0-P6 implementation sequence and rollback
Verification_Test_Plans.md	Golden, security, resilience, regression plans
Competitor_Analysis.md / Competitor_Matrix.csv	Public-market design loop; no proprietary-internals claims
Sources/source_ledger.csv + Sources/HuggingFace	Primary URLs and immutable saved source evidence
QA/final_architecture_freeze_audit.csv	Final architecture-freeze verification
QA/package_validation.json	Structured consistency, portability and preservation validation

Terminology: specialist = evidence producer; Gateway = deterministic policy owner; candidate = not yet scientifically promoted; shadow = no decision authority; private = controlled HF repository/Endpoint after creation and qualification.